

ALADIN EEG Validation Suite – S02-DMT

30 Research-Grade Plots + Summary

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Summary Statistics

ALADIN S02-DMT Validation Summary

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PAC (Tort MI): 0.0000

$\Delta\alpha$ pre $t \approx 22.132812$: 4.1348

$\Delta\alpha$ post $t \approx 22.132812$: 3.0457

$\Delta\alpha$ drop %: 26.3%

Permutation p $\Delta\alpha$: 0.0462

Permutation p Gamma: 0.0000

Cohen's d $\Delta\alpha$: 0.653

Cohen's d Gamma: 1.817

Granger raw p-values: [np.float64(0.7291636746633675), np.float64(1.2639238103932e-08)]

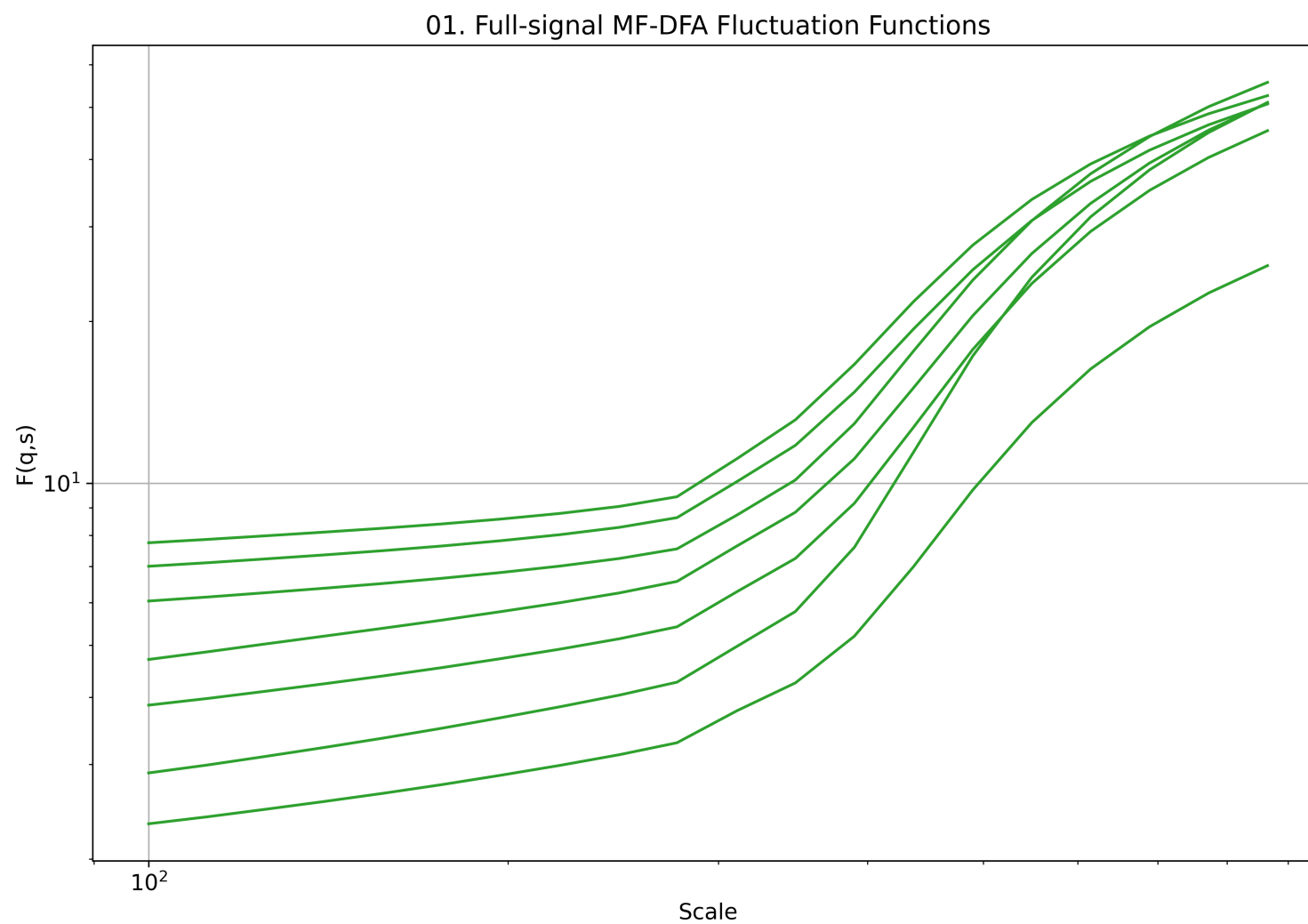
Granger FDR corrected p-values: [0.7291636746633675, 1.5799047629915427e-08, 1.5799047629915427e-08]

$\Delta\alpha$ at t_{peak} : 2.1222

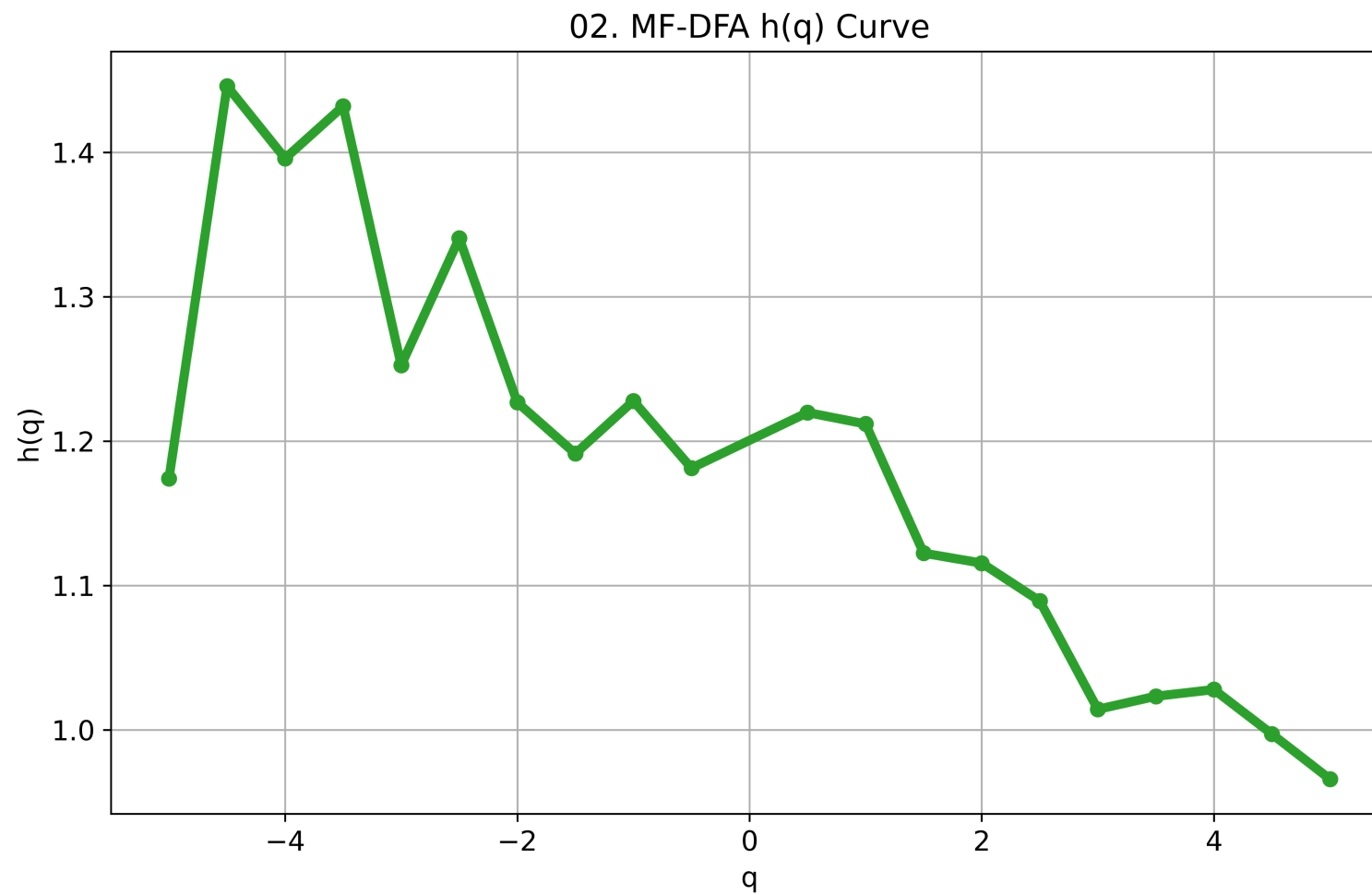
$\Delta\alpha$ min: 0.9986

$\Delta\alpha$ max: 9.9149

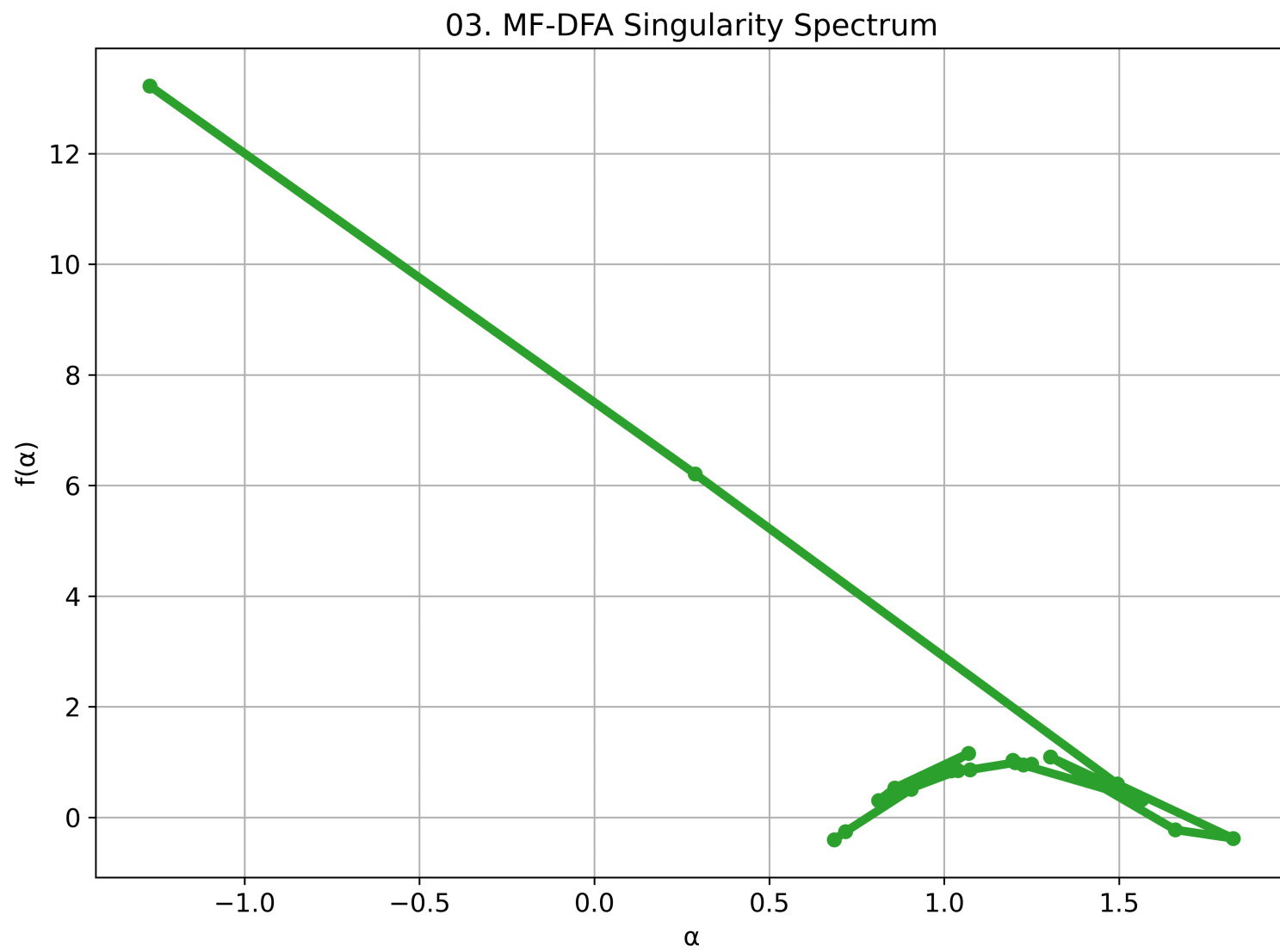
Z-score at t_{peak} : -1.12



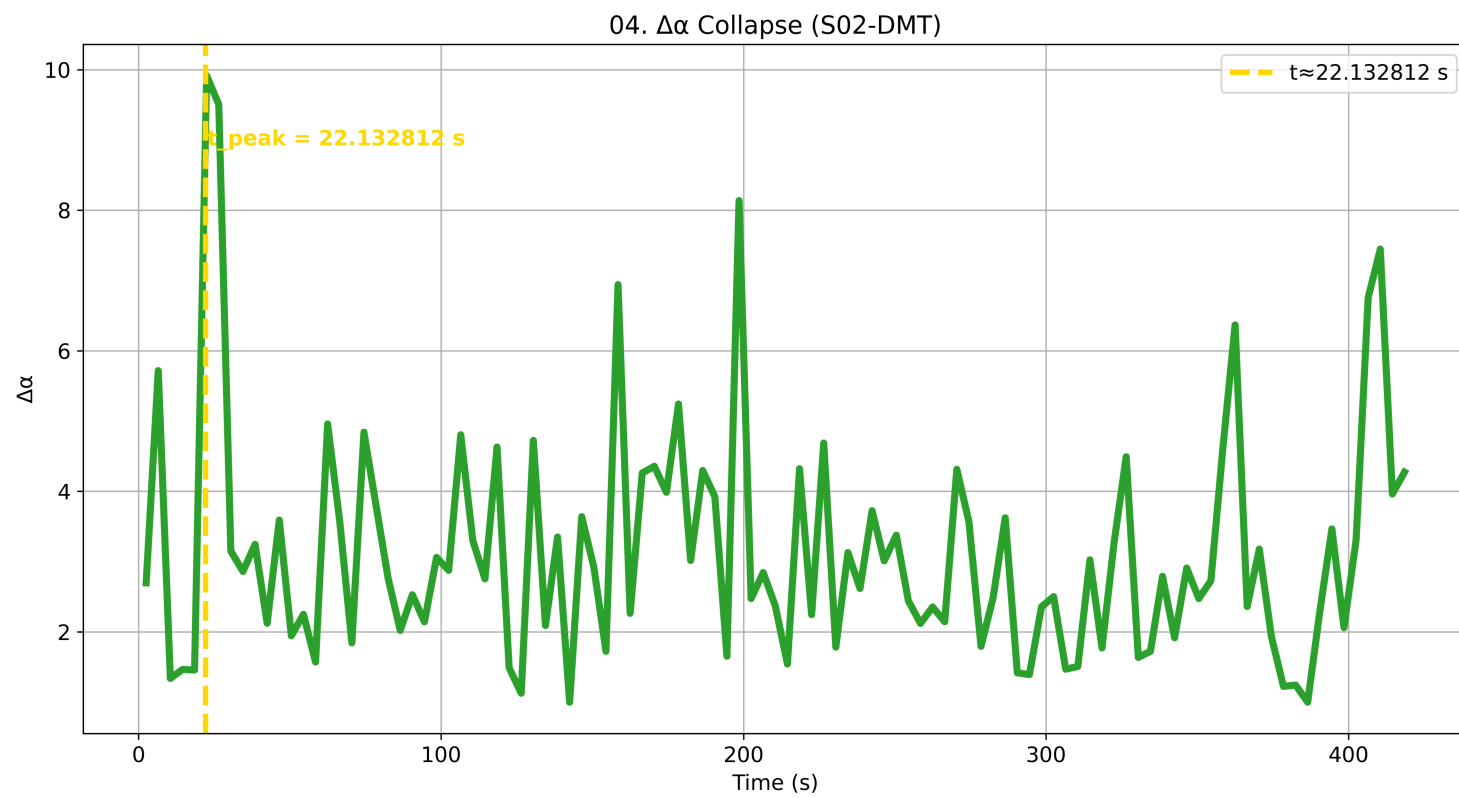
Full-signal MF-DFA fluctuation functions across q orders ($t_{\text{peak}} \approx 22.132812$ s)



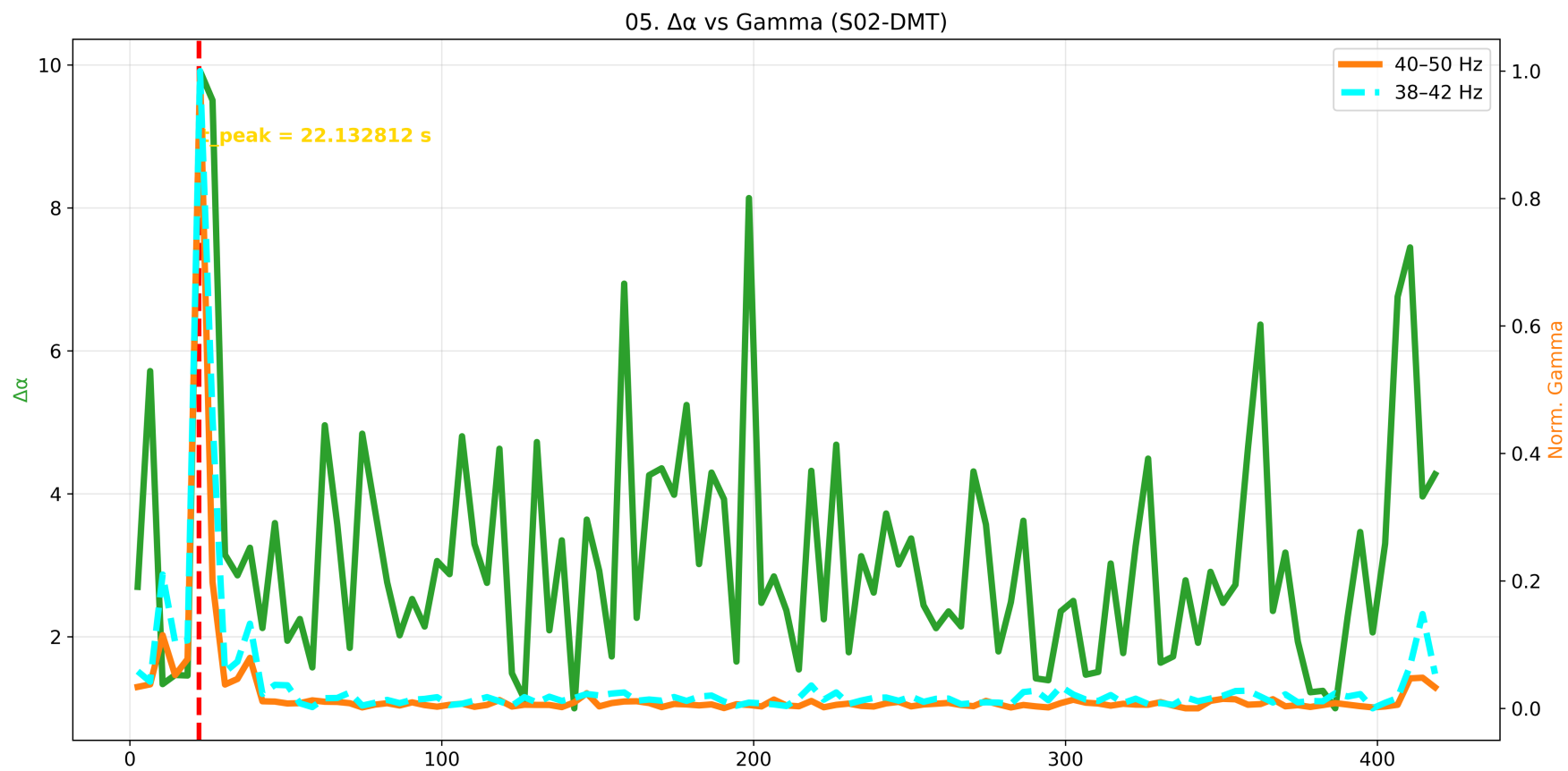
MF-DFA generalized Hurst exponent $h(q)$ curve showing multifractality ($t_{\text{peak}} \approx 22.132812$ s)



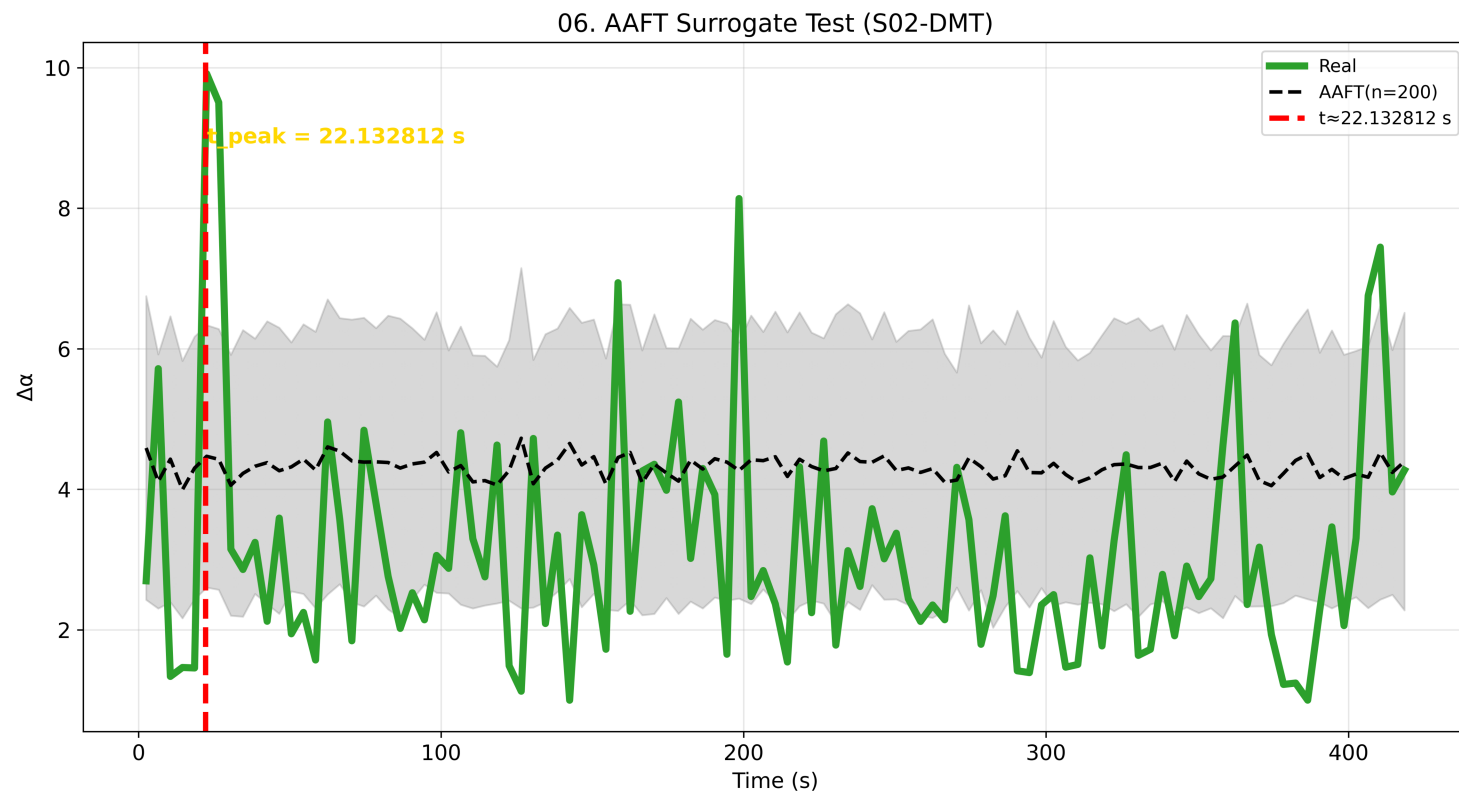
MF-DFA singularity spectrum $f(\alpha)$ – width indicates multifractal strength ($t_{\text{peak}} \approx 22.132812$ s)



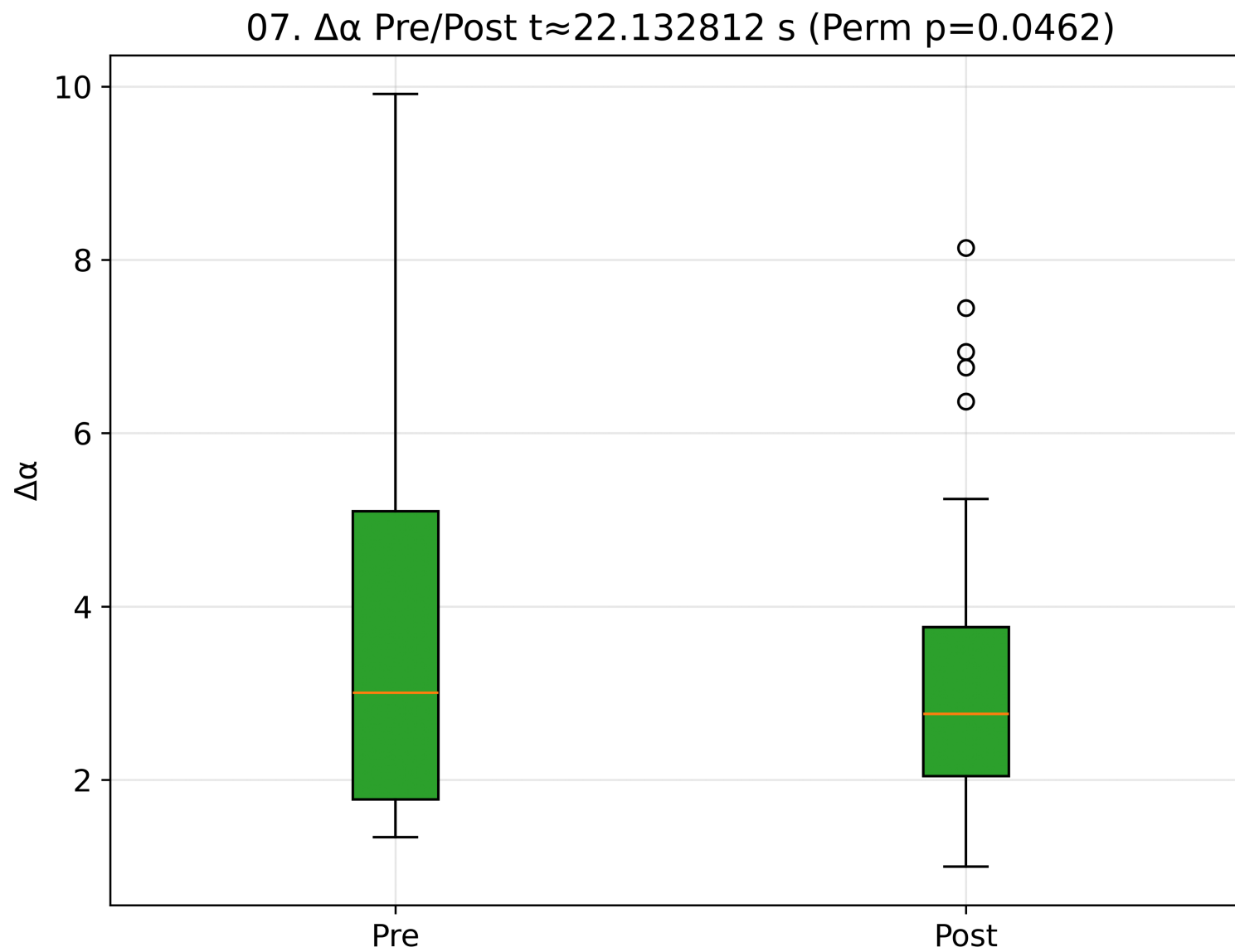
Time-resolved $\Delta\alpha$ collapse with objective peak at 22.132812 s



$\Delta\alpha$ collapse aligned with 40-50 Hz gamma power (peak at 22.132812 s)

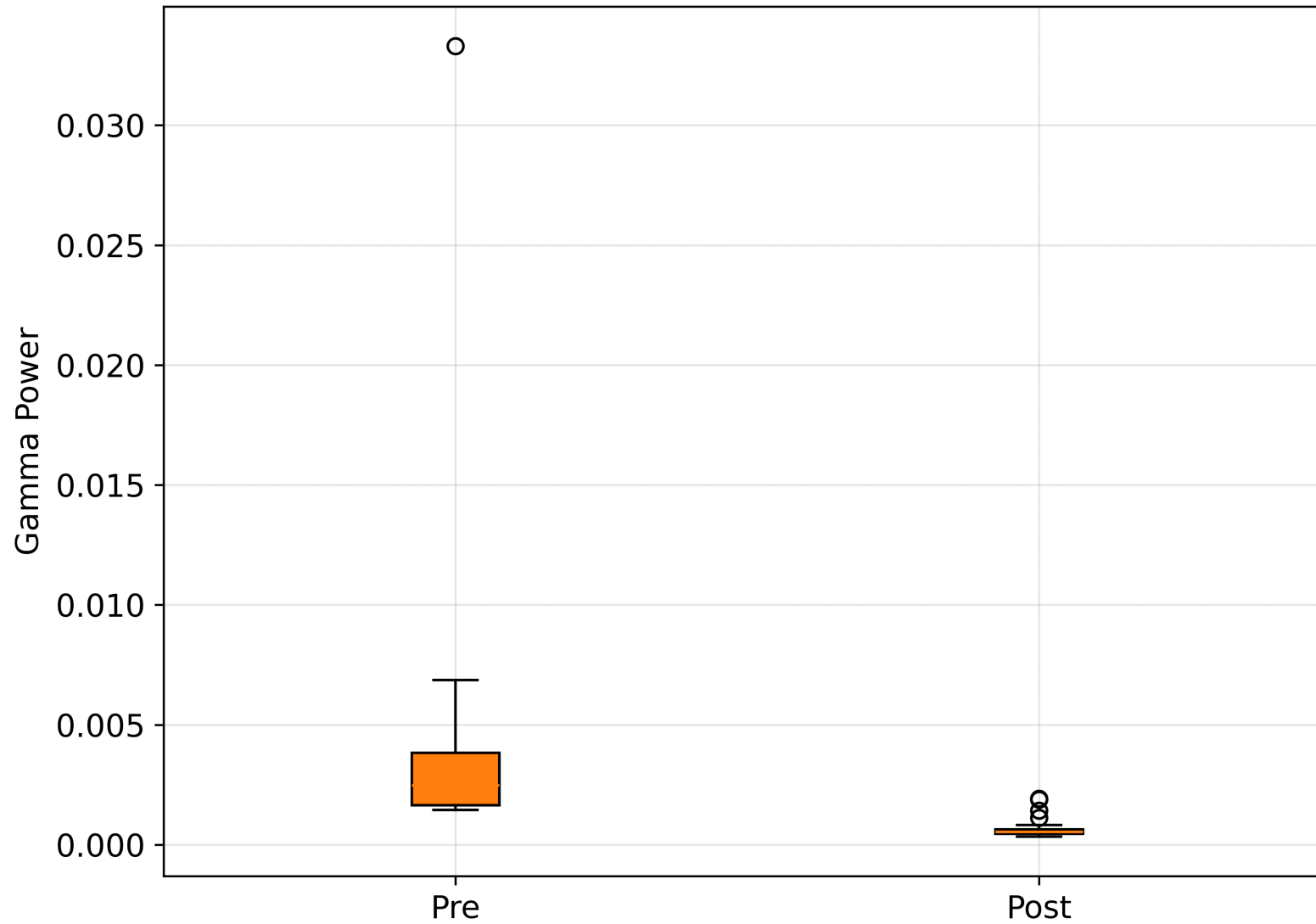


AAFT surrogate test with objective peak at 22.132812 s

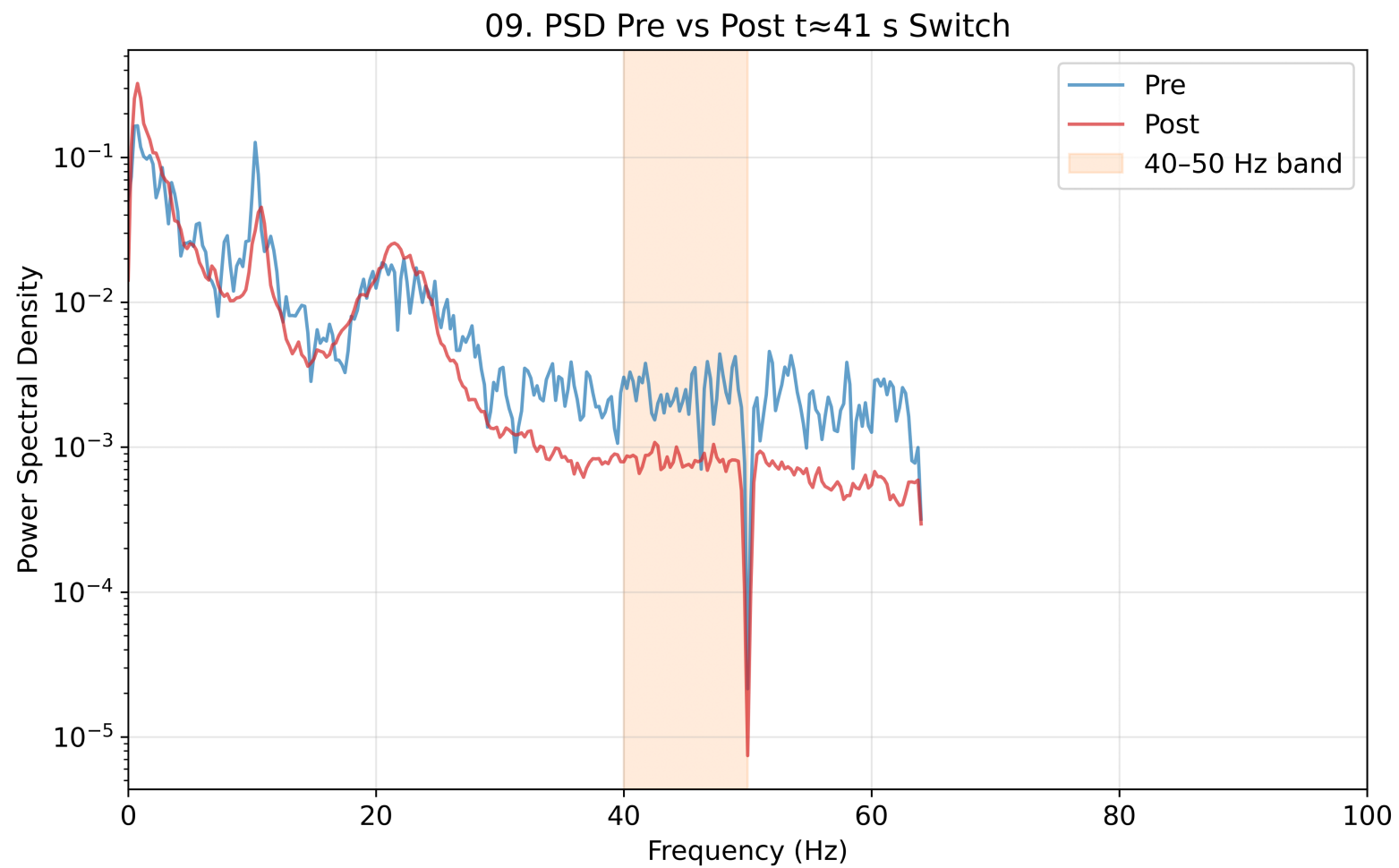


$\Delta\alpha$ pre vs post $t \approx 22.132812$ s with permutation test

08. Gamma 40-50 Hz Pre/Post $t \approx 22.132812$ s (Perm $p=0.0000$)

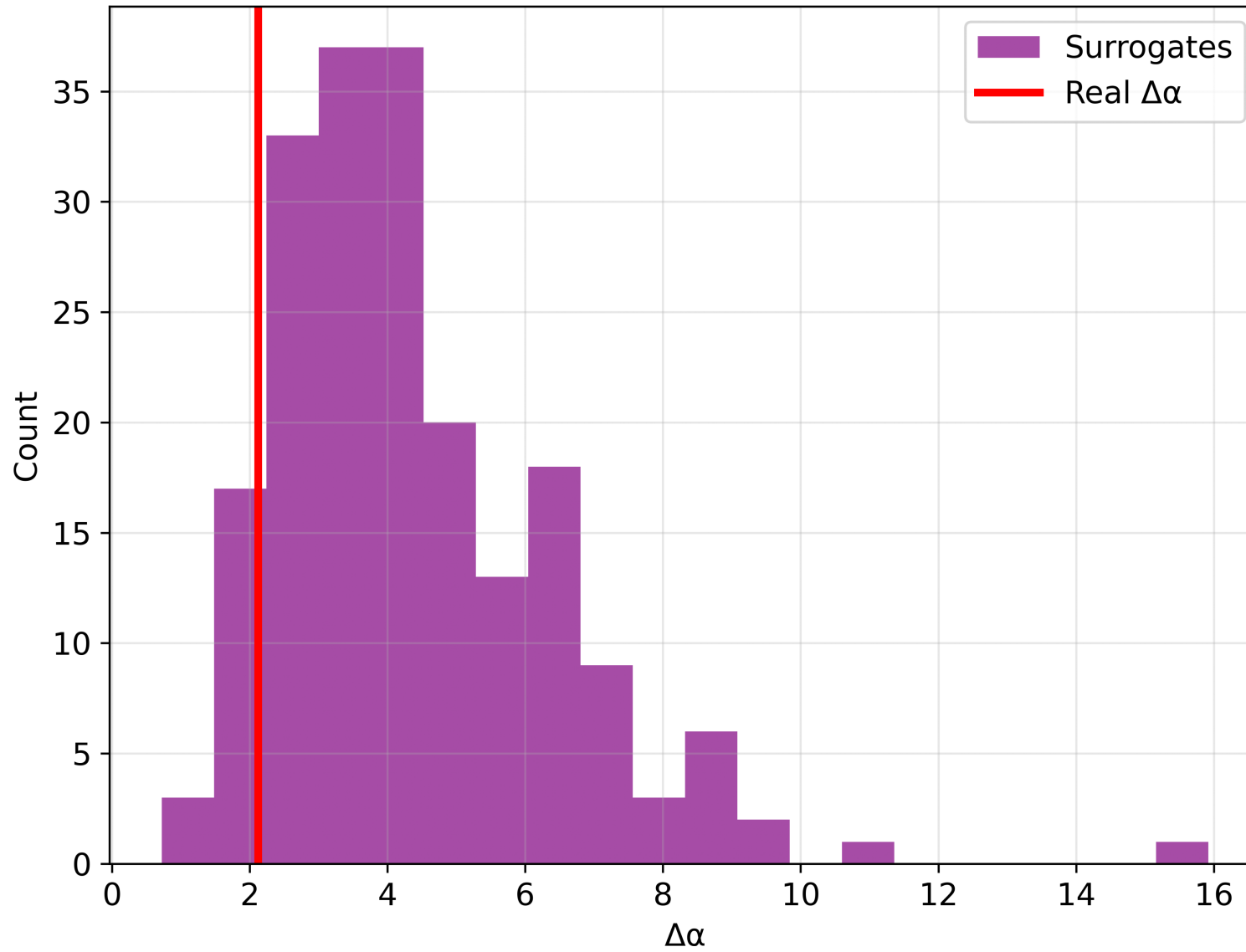


Gamma 40-50 Hz power pre vs post $t \approx 22.132812$ s with permutation test



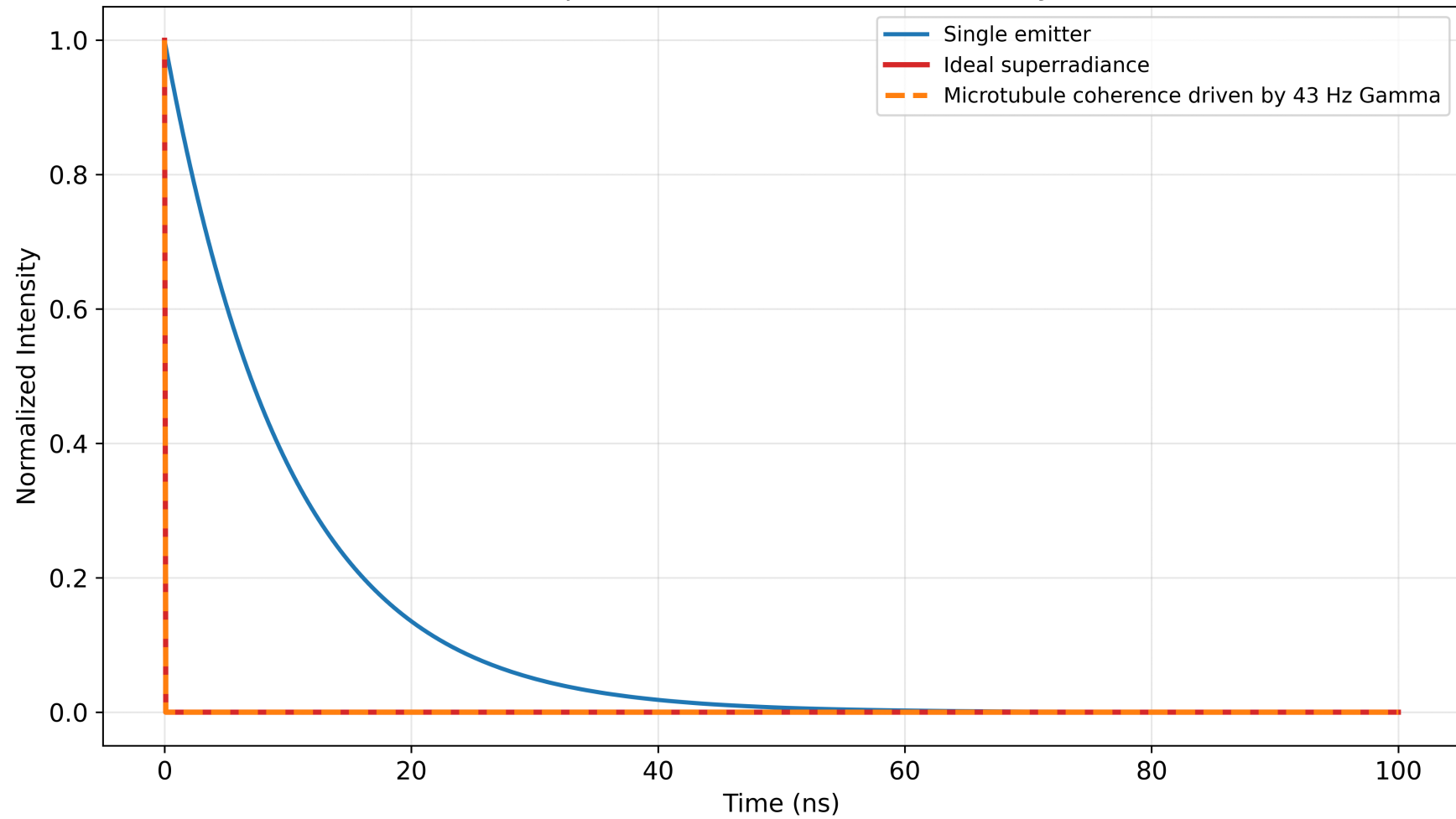
Power spectral density pre vs post $t \approx 22.132812$ s switch

10. Surrogate Z-score Distribution at $t \approx 41$ s

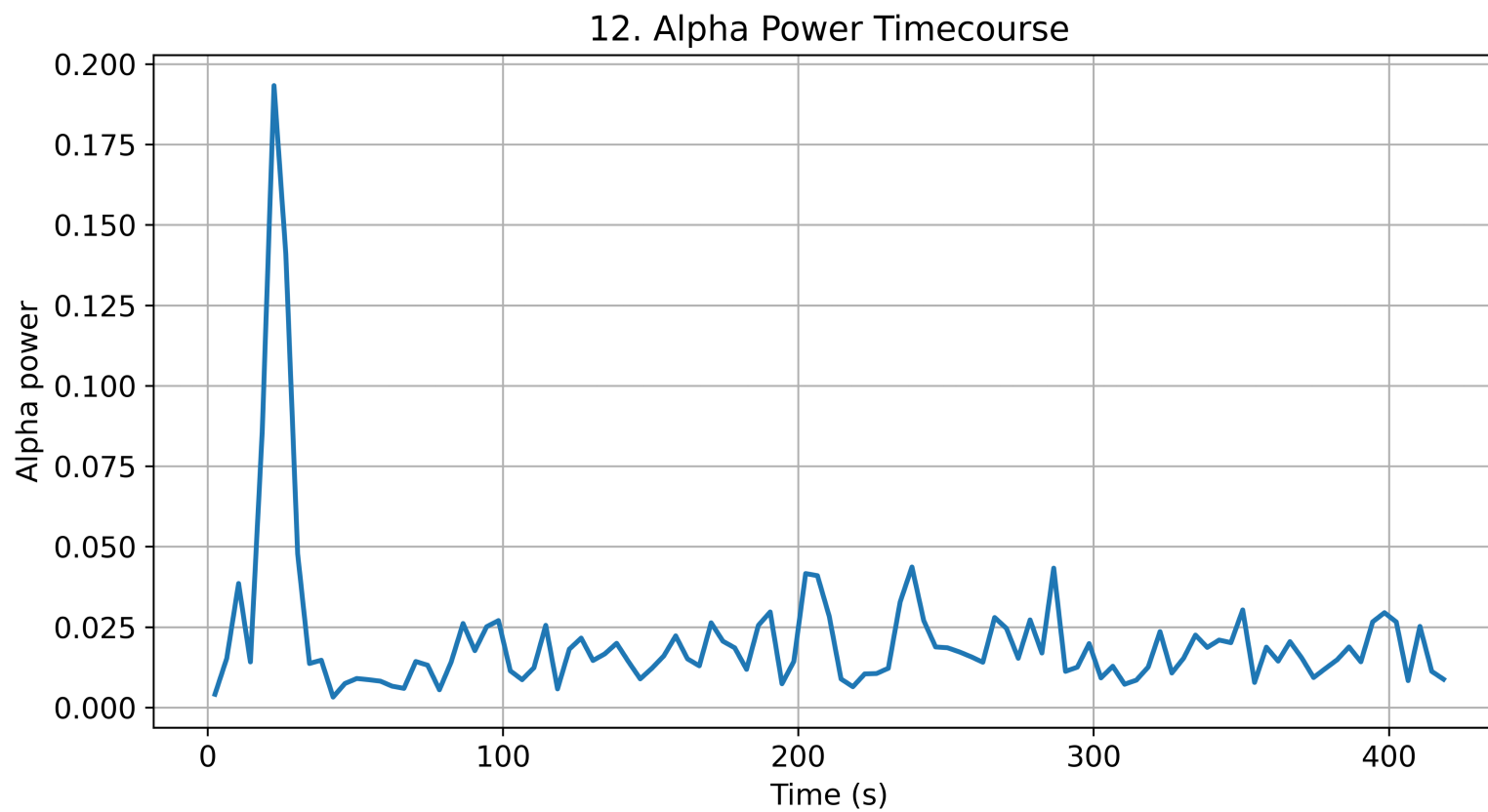


Surrogate distribution of $\Delta\alpha$ at peak time (real value marked, $t_{\text{peak}} \approx 22.132812$ s)

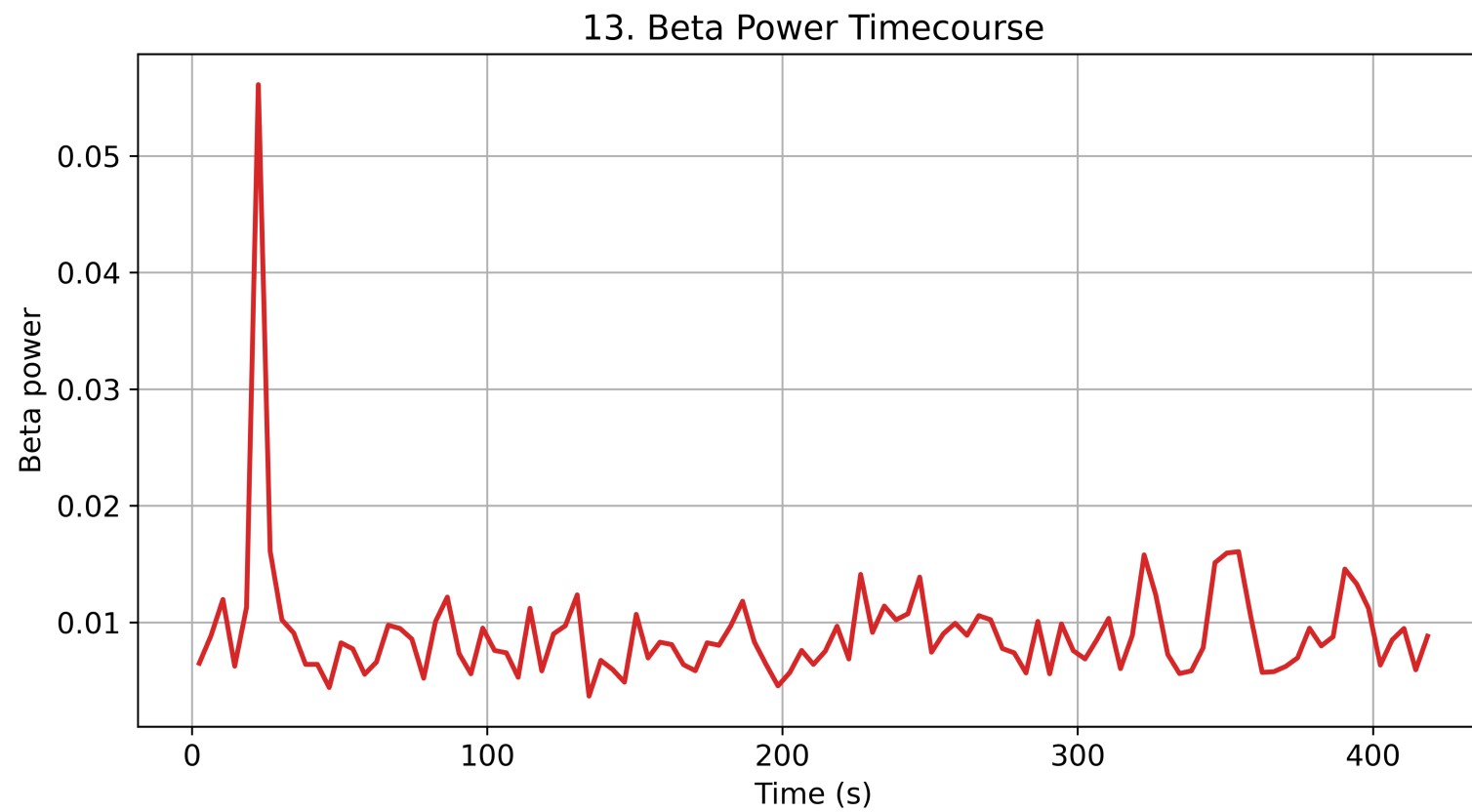
11. Microtubule Superradiance Coherence Driven by 43 Hz Gamma



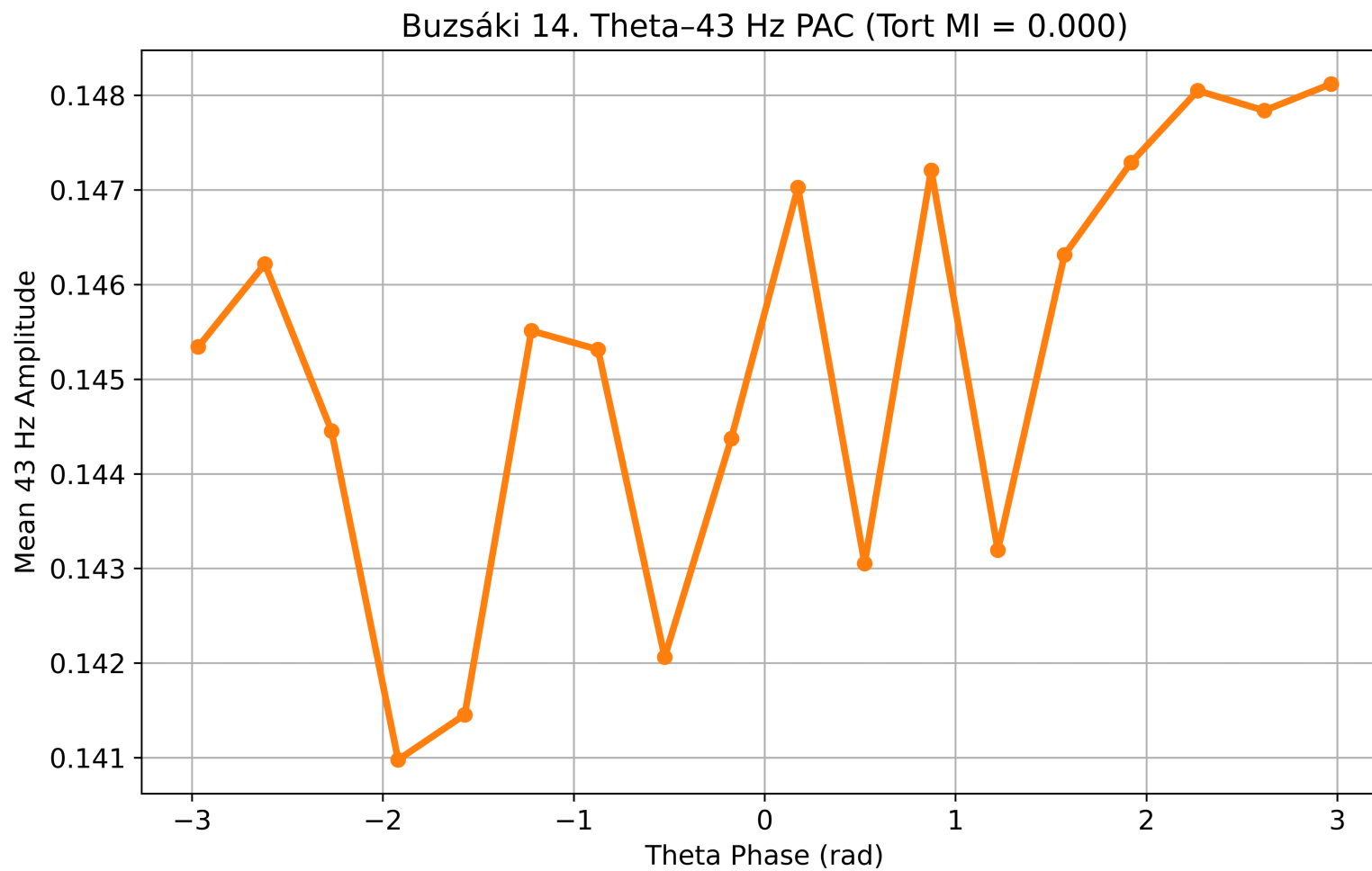
Microtubule Superradiance Coherence Driven by 43 Hz Gamma ($t_{\text{peak}} \approx 22.132812$ s)



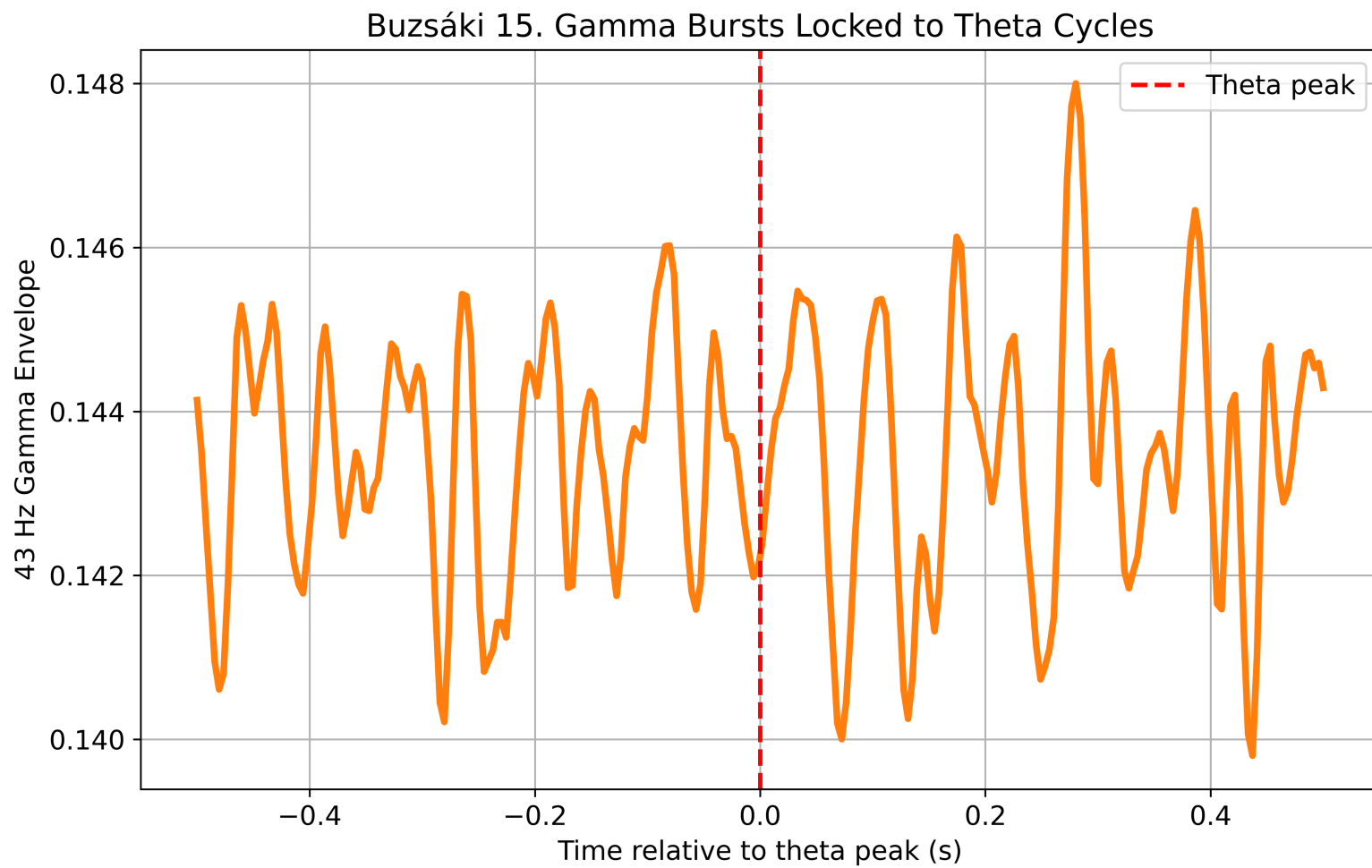
Alpha (8-12 Hz) power timecourse ($t_{\text{peak}} \approx 22.132812$ s)



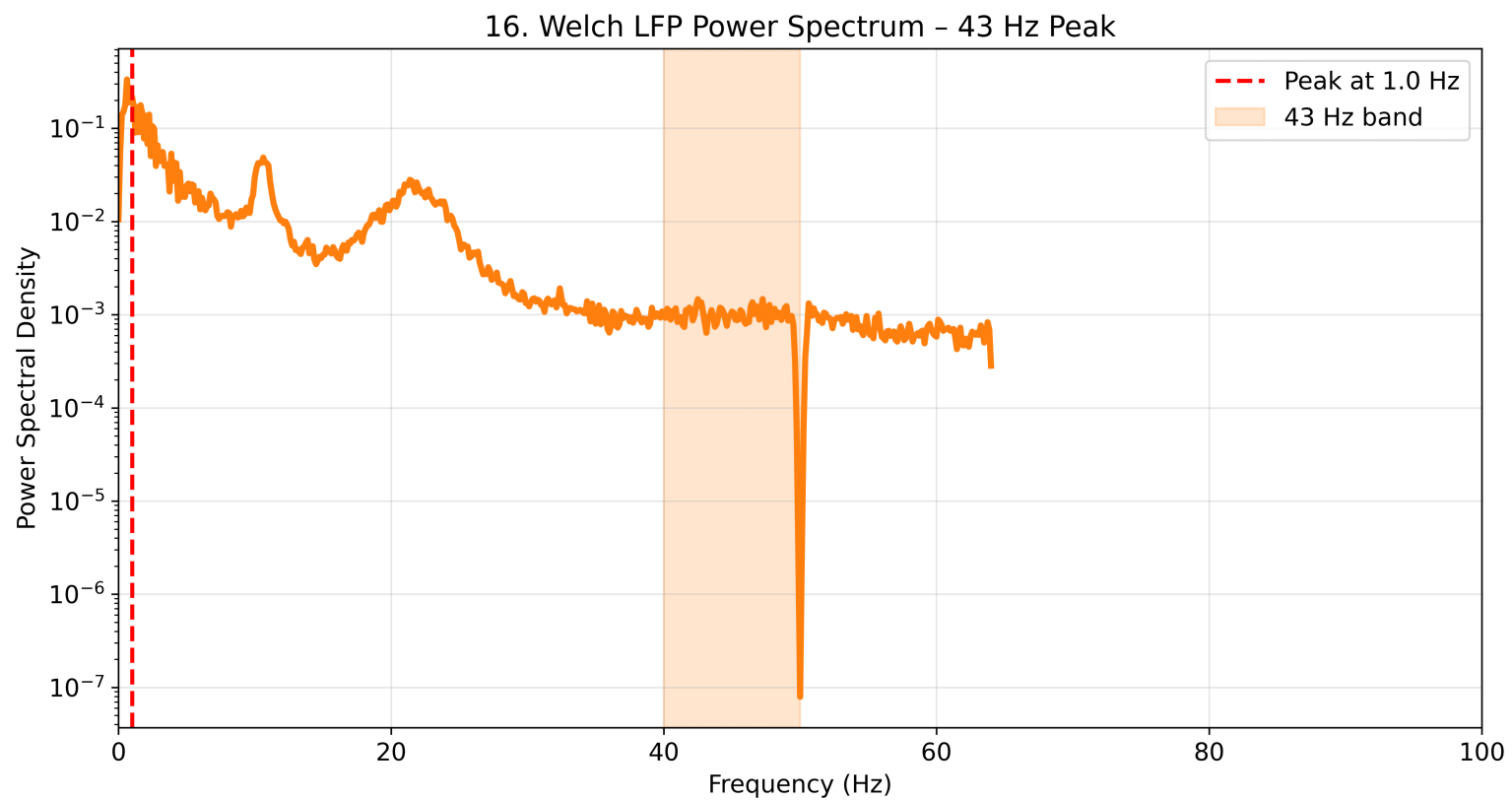
Beta (12-30 Hz) power timecourse ($t_{\text{peak}} \approx 22.132812$ s)



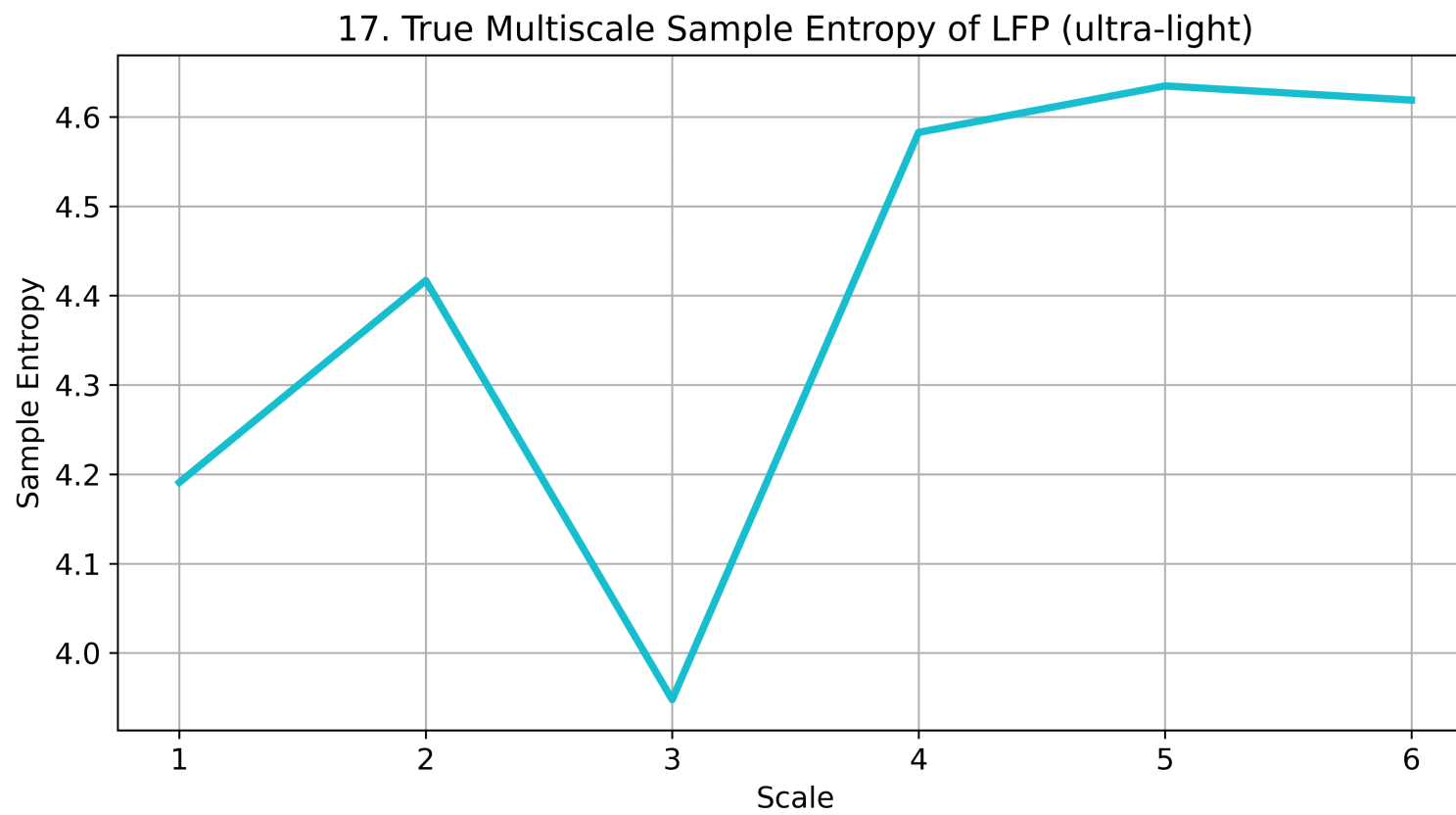
Buzsáki-style theta-43 Hz phase-amplitude coupling (Tort MI) ($t_{\text{peak}} \approx 22.132812$ s)



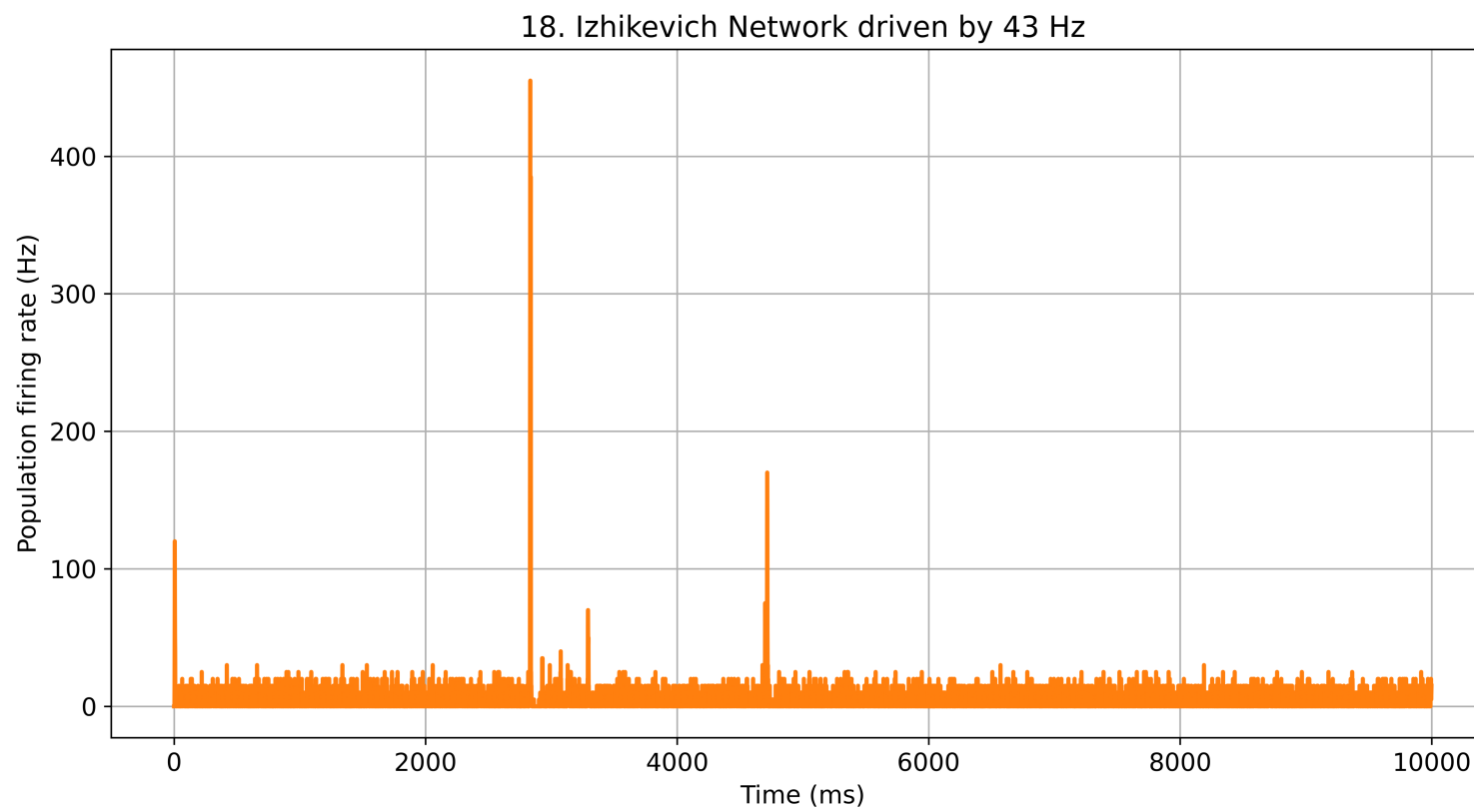
Gamma bursts locked to theta cycles (Buzsáki canonical) ($t_{\text{peak}} \approx 22.132812$ s)



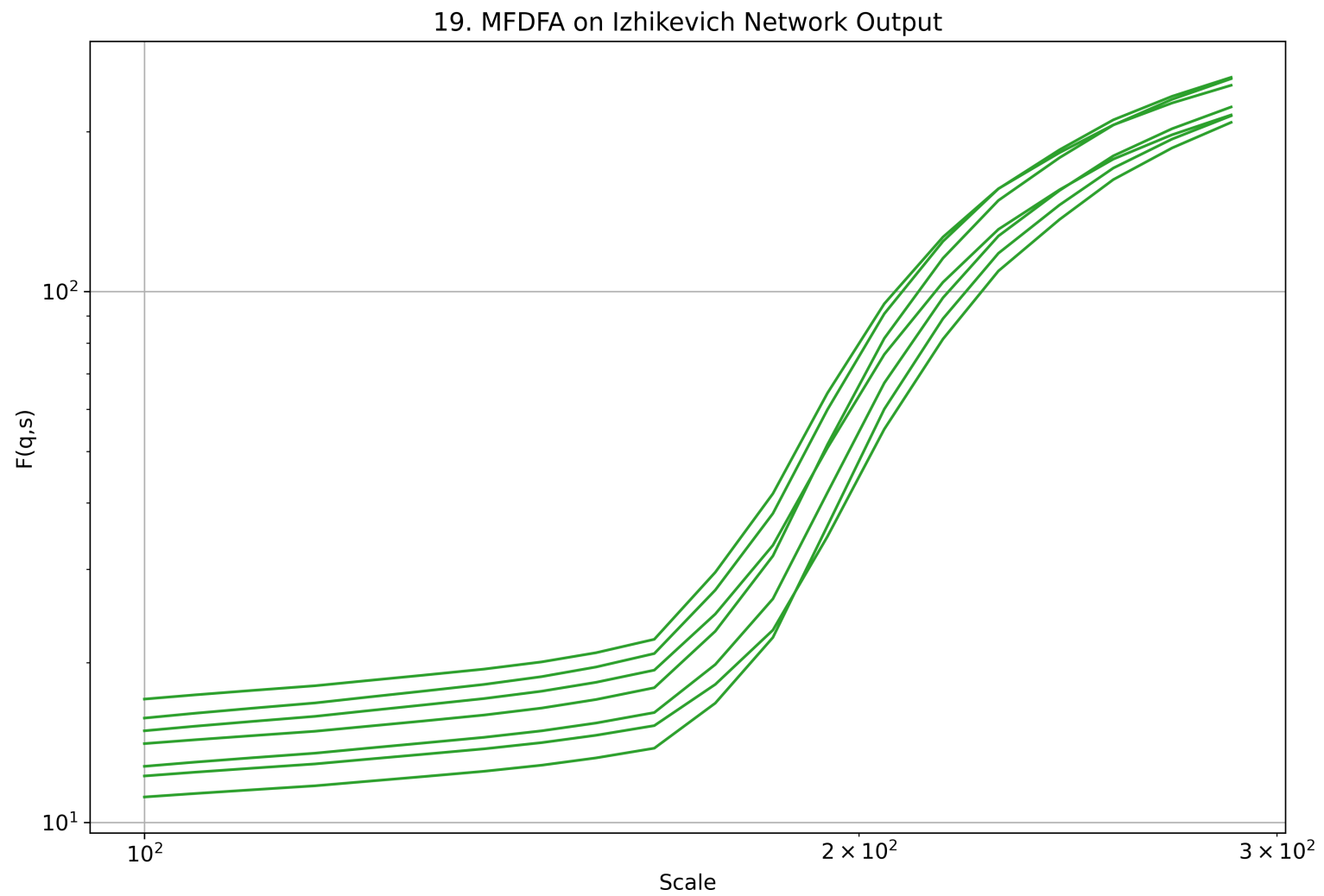
Welch LFP power spectrum showing 43 Hz peak ($t_{\text{peak}} \approx 22.132812$ s)



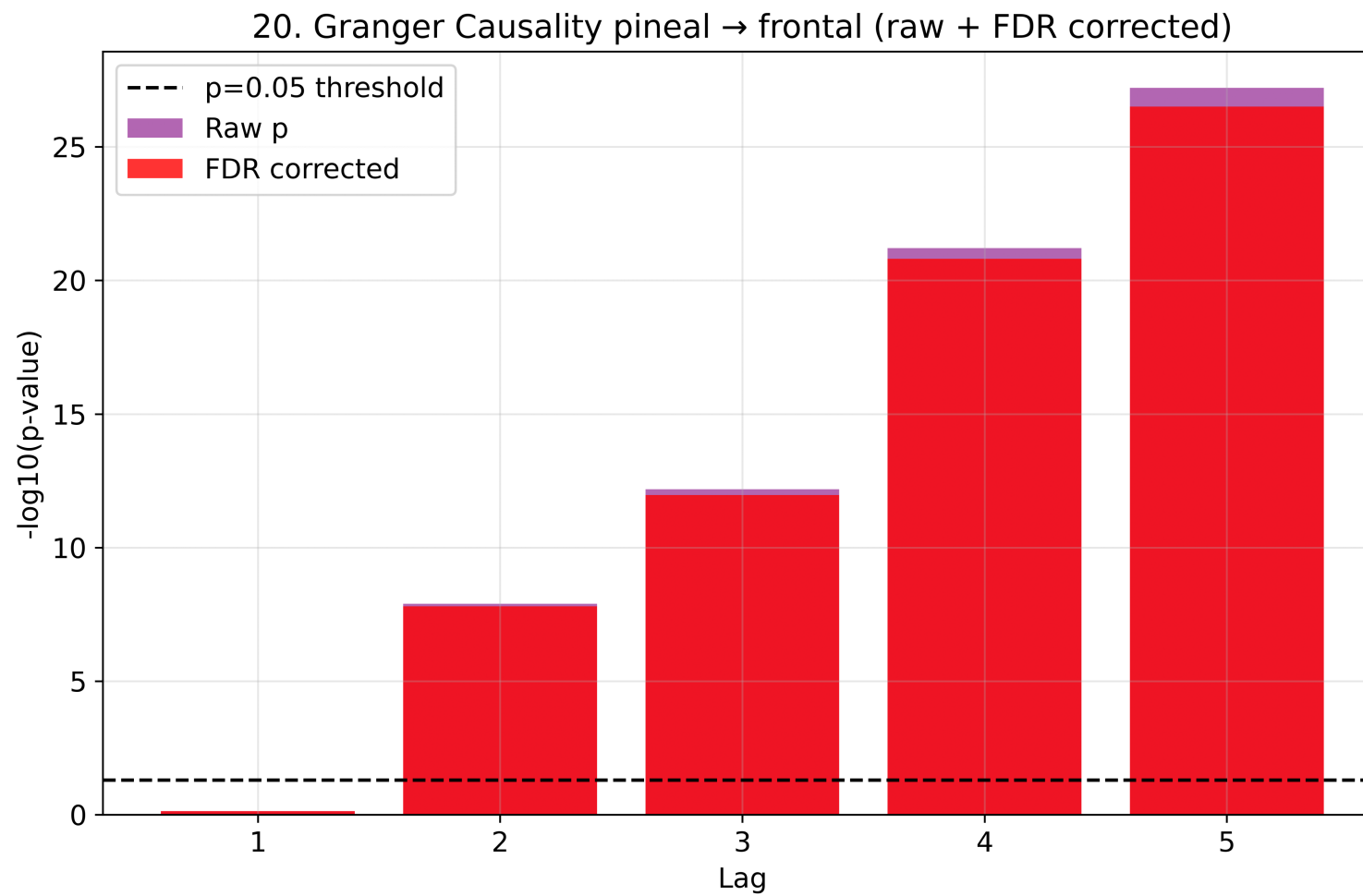
True multiscale sample entropy of LFP signal ($t_{\text{peak}} \approx 22.132812$ s)



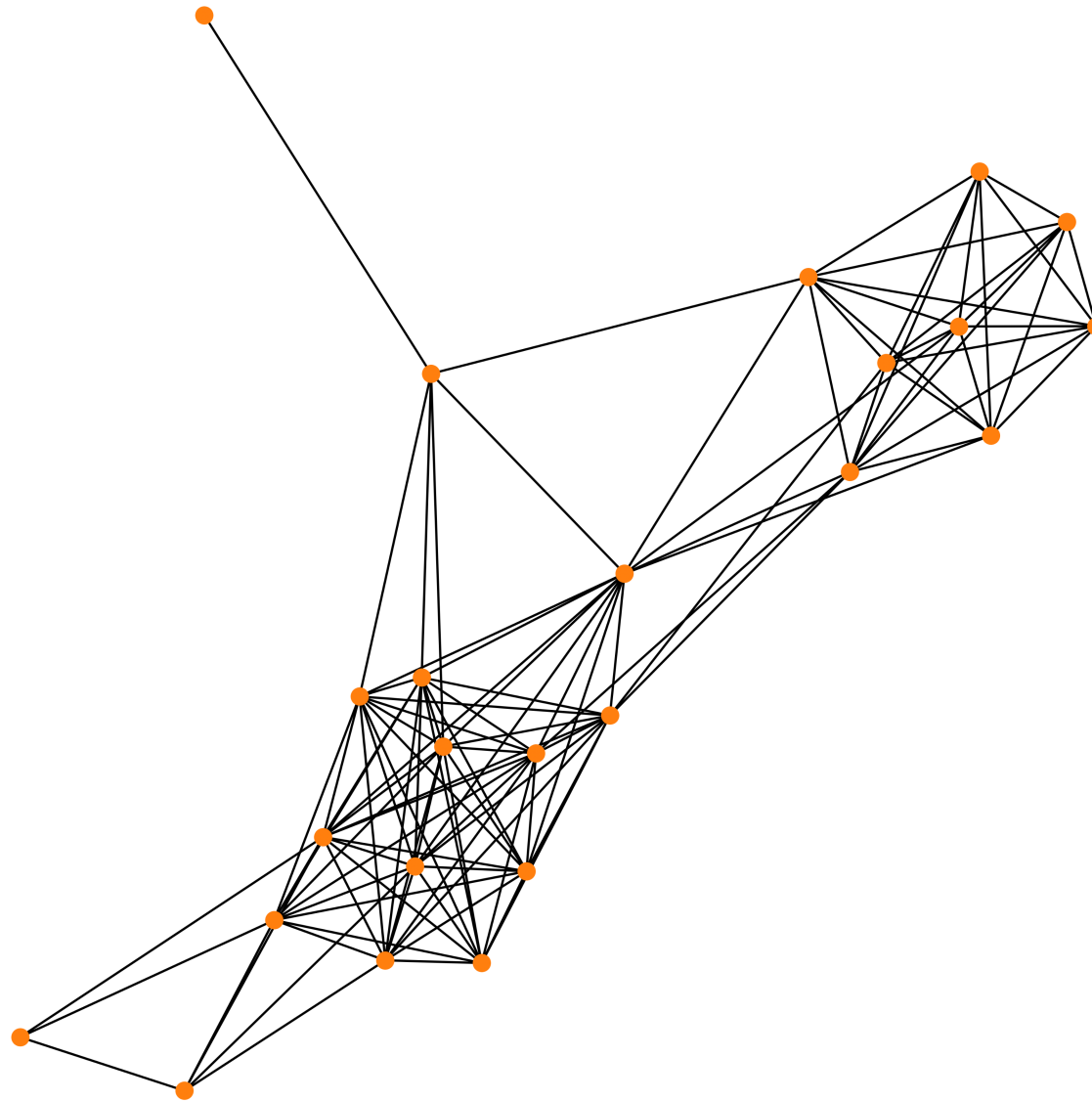
Izhikevich spiking network driven by observed 43 Hz gamma ($t_{\text{peak}} \approx 22.132812$ s)



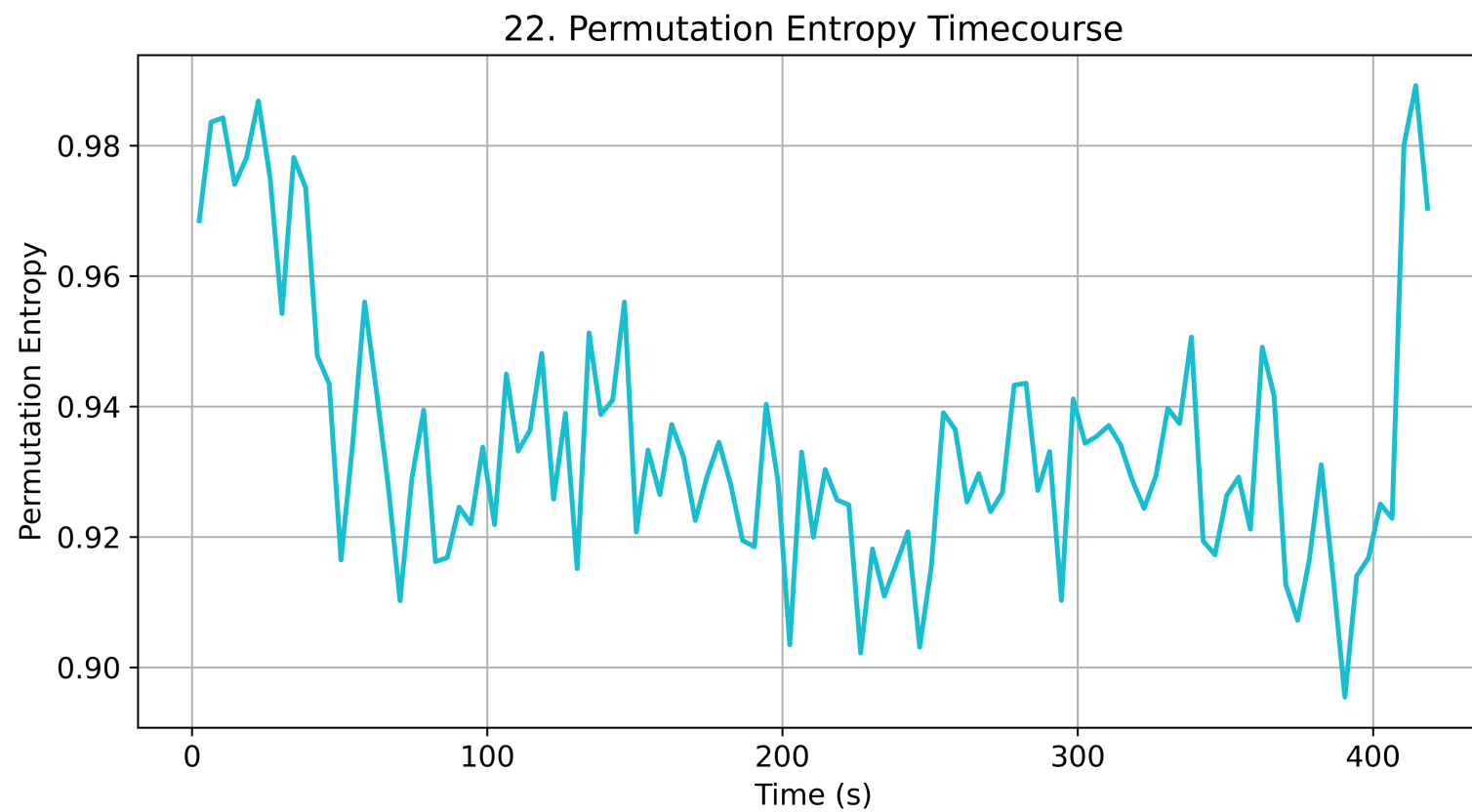
MF-DFA applied to Izhikevich network output ($t_{\text{peak}} \approx 22.132812$ s)



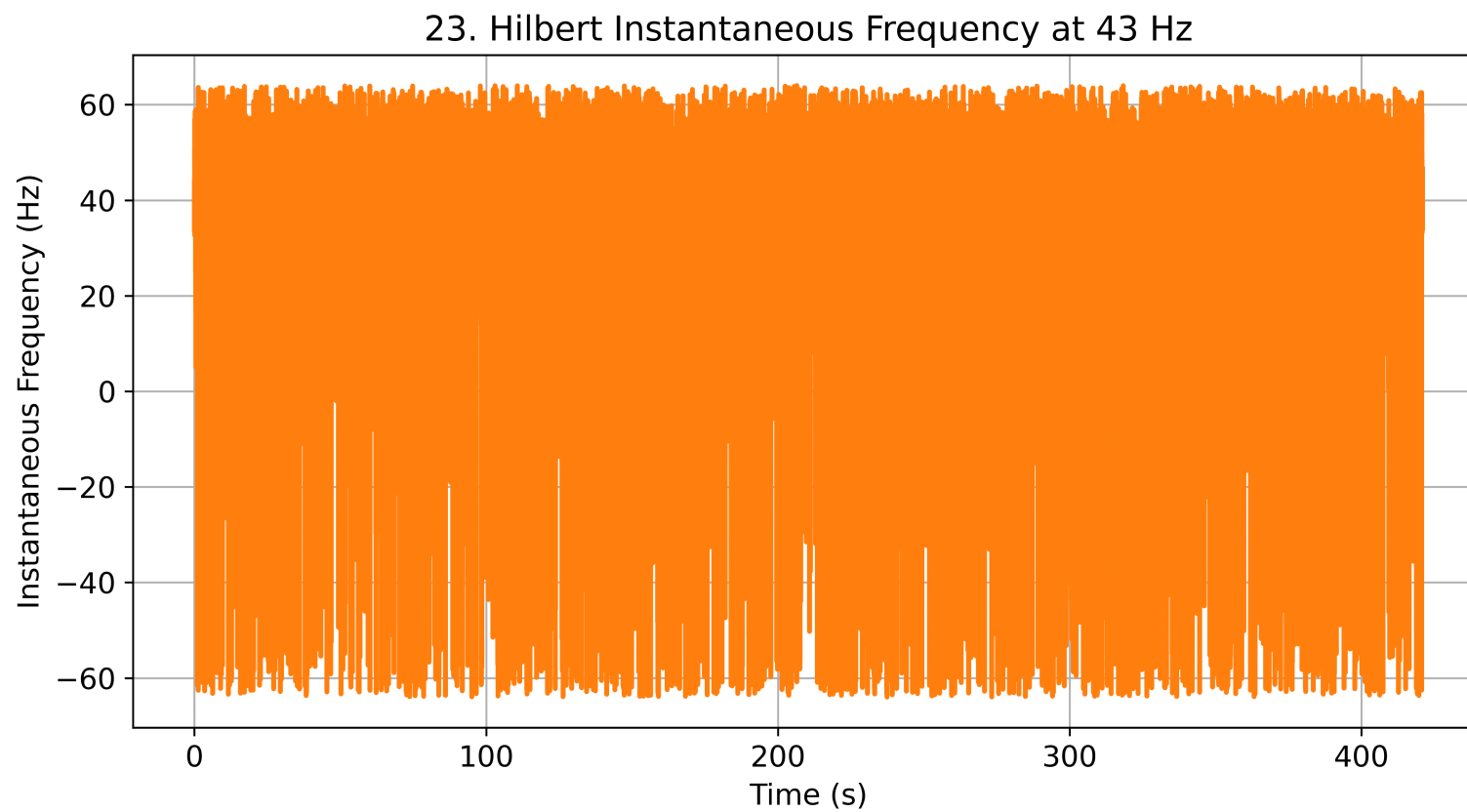
Granger causality pineal → frontal with FDR correction ($t_{\text{peak}} \approx 22.132812$ s)



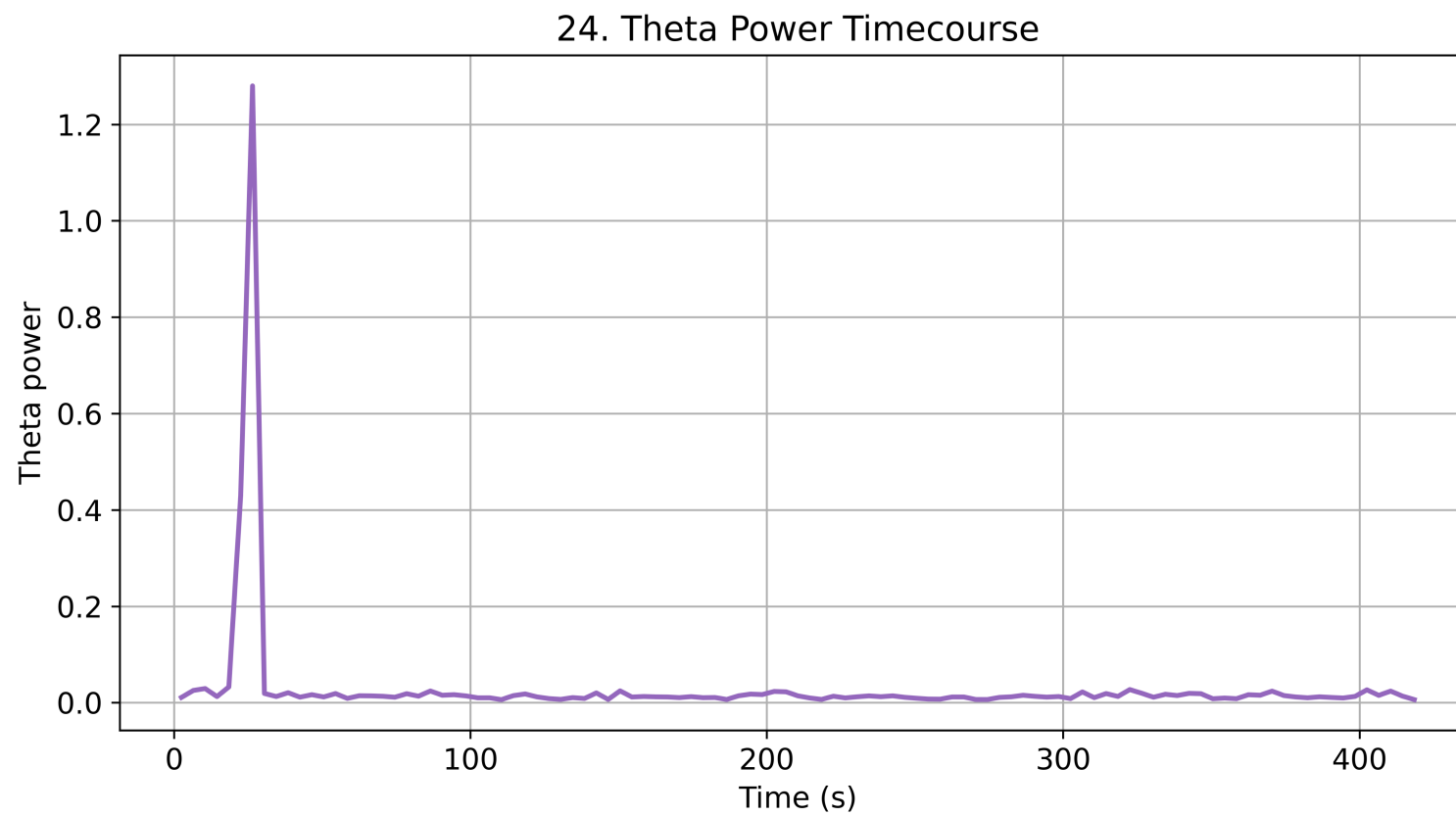
Correlation-based EEG connectivity graph (no self-loops) ($t_{\text{peak}} \approx 22.132812$ s)



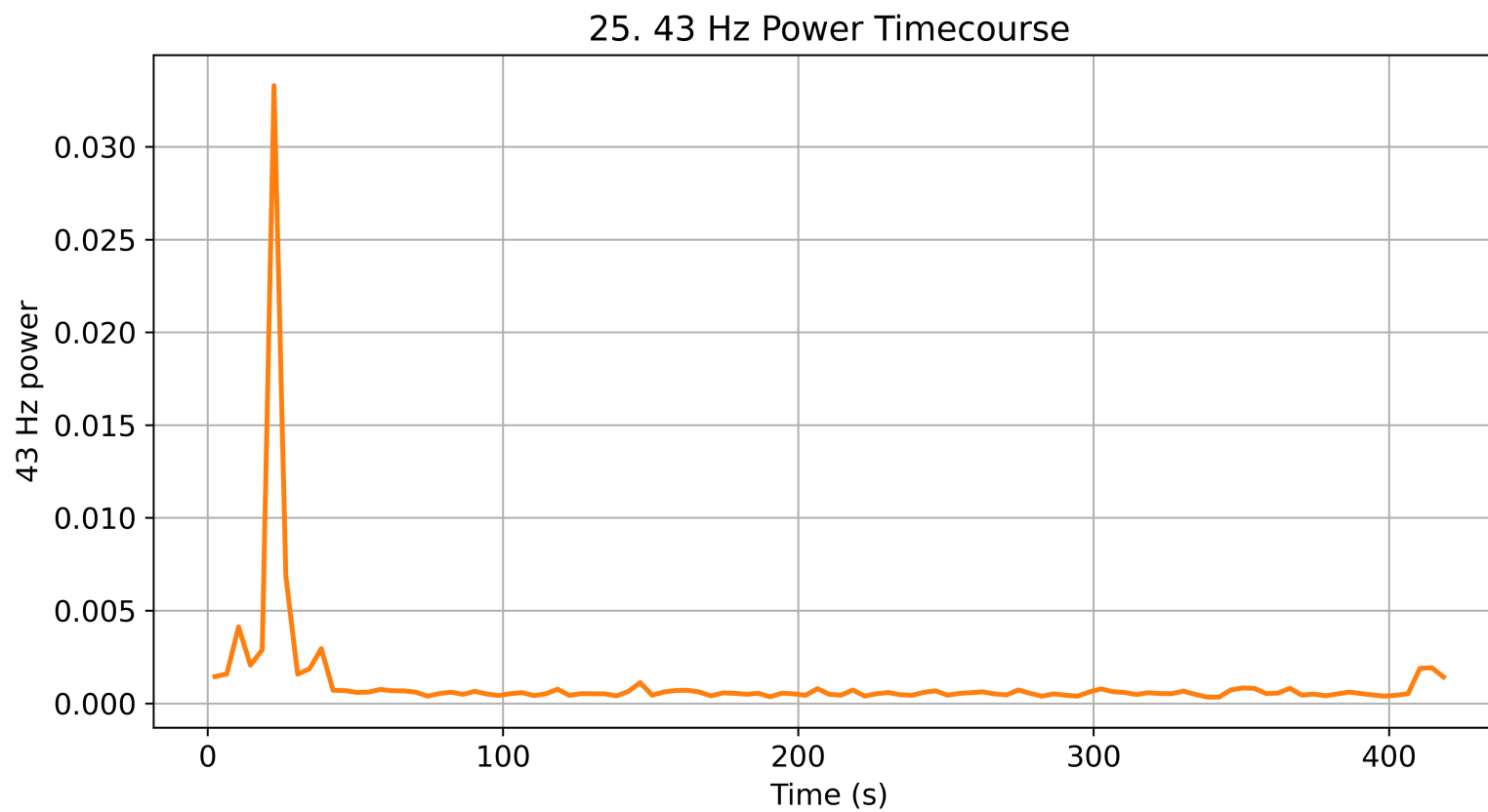
Permutation entropy timecourse ($t_{\text{peak}} \approx 22.132812$ s)



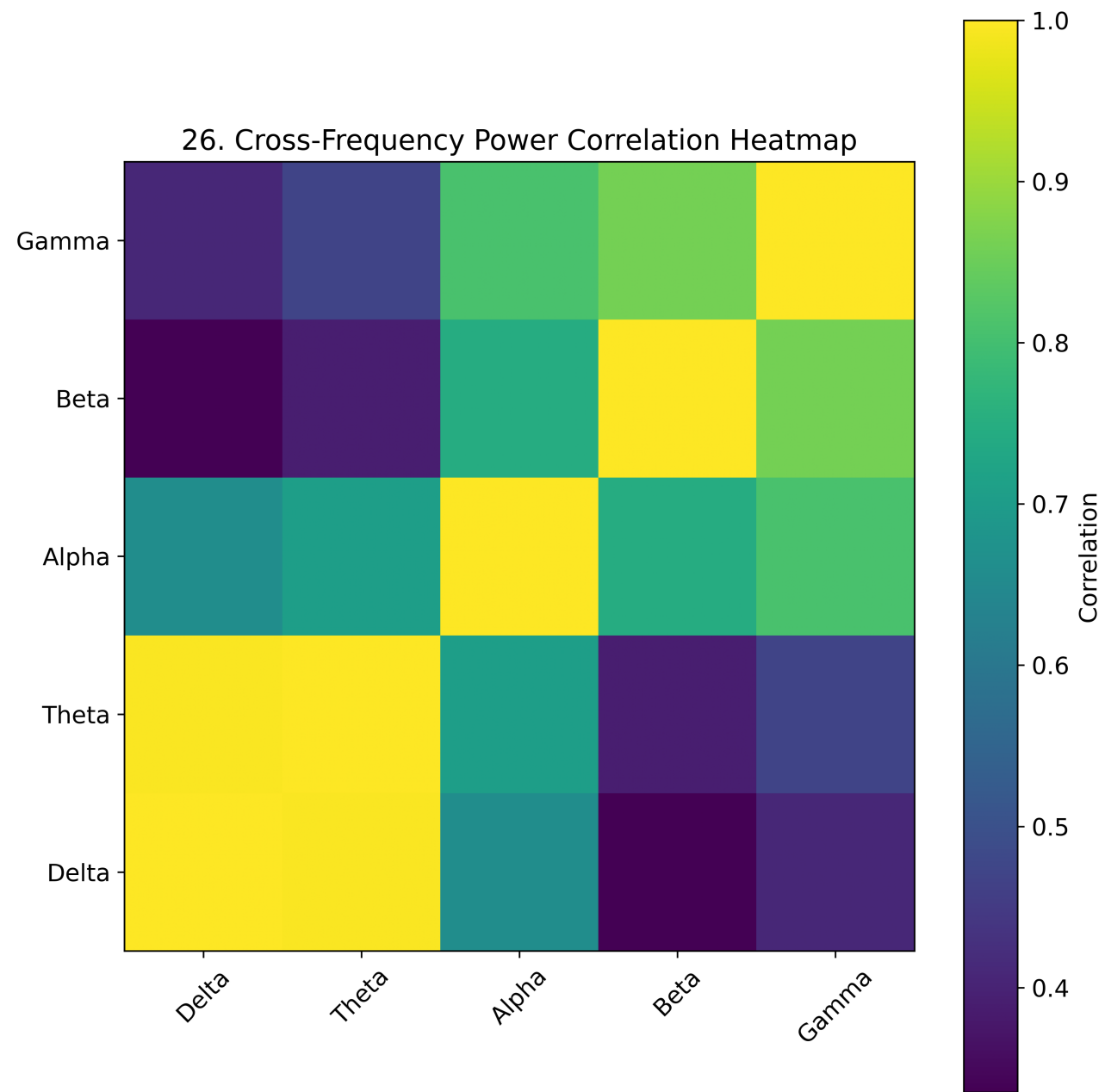
Instantaneous frequency of 43 Hz gamma component ($t_{\text{peak}} \approx 22.132812$ s)



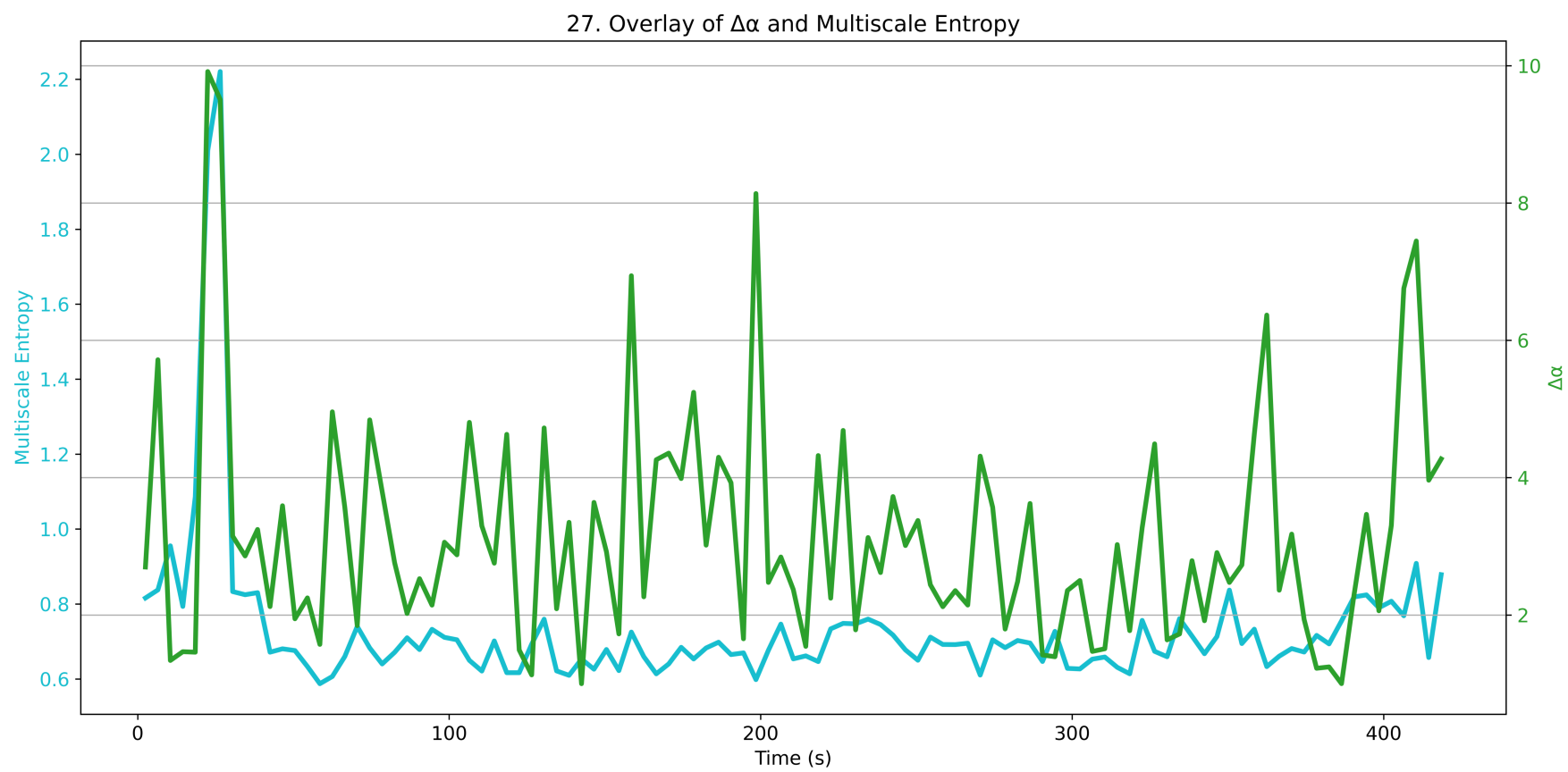
Theta (4-8 Hz) power timecourse ($t_{\text{peak}} \approx 22.132812$ s)



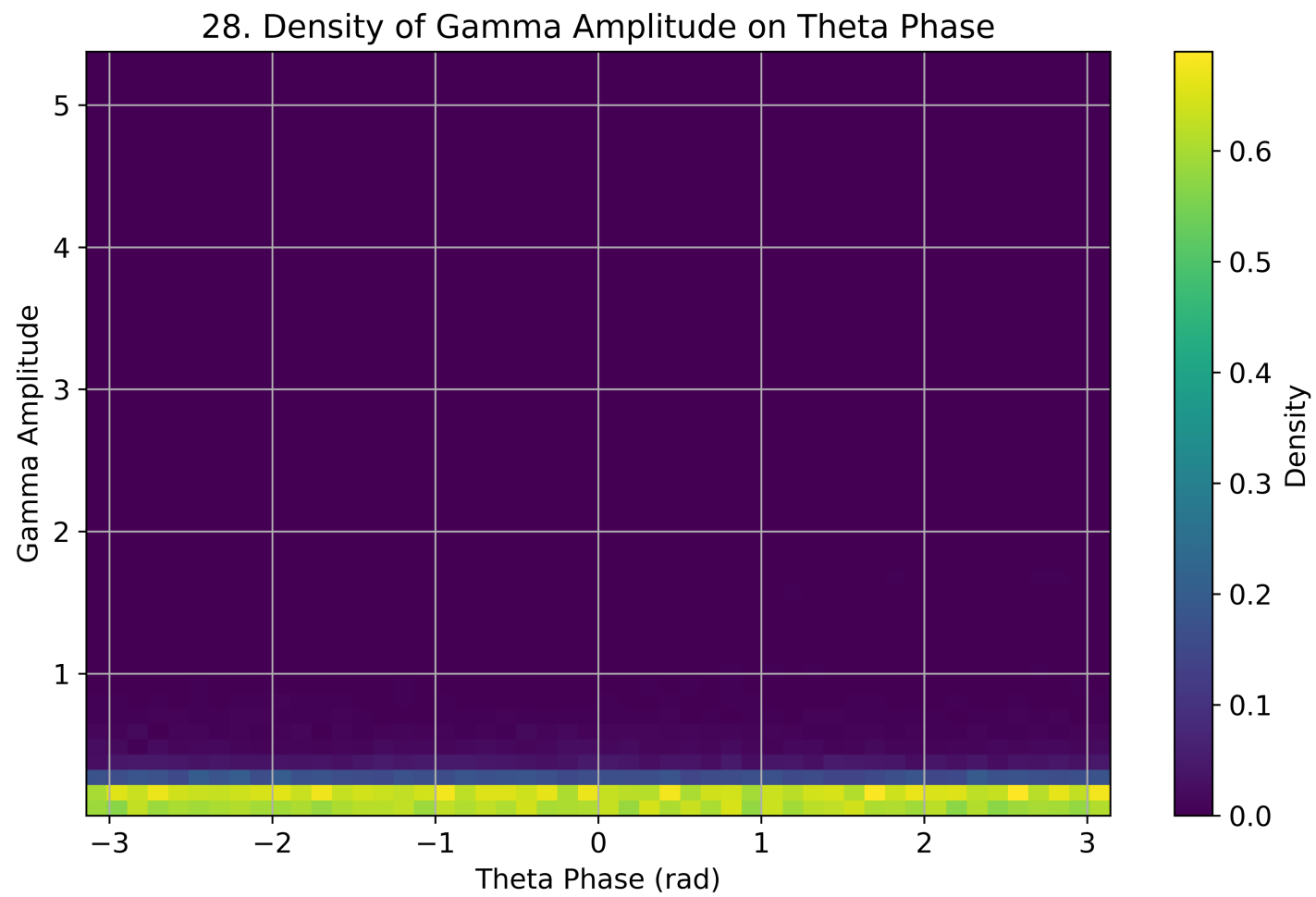
43 Hz gamma power timecourse ($t_{\text{peak}} \approx 22.132812$ s)



Cross-frequency power correlation heatmap ($t_{\text{peak}} \approx 22.132812$ s)

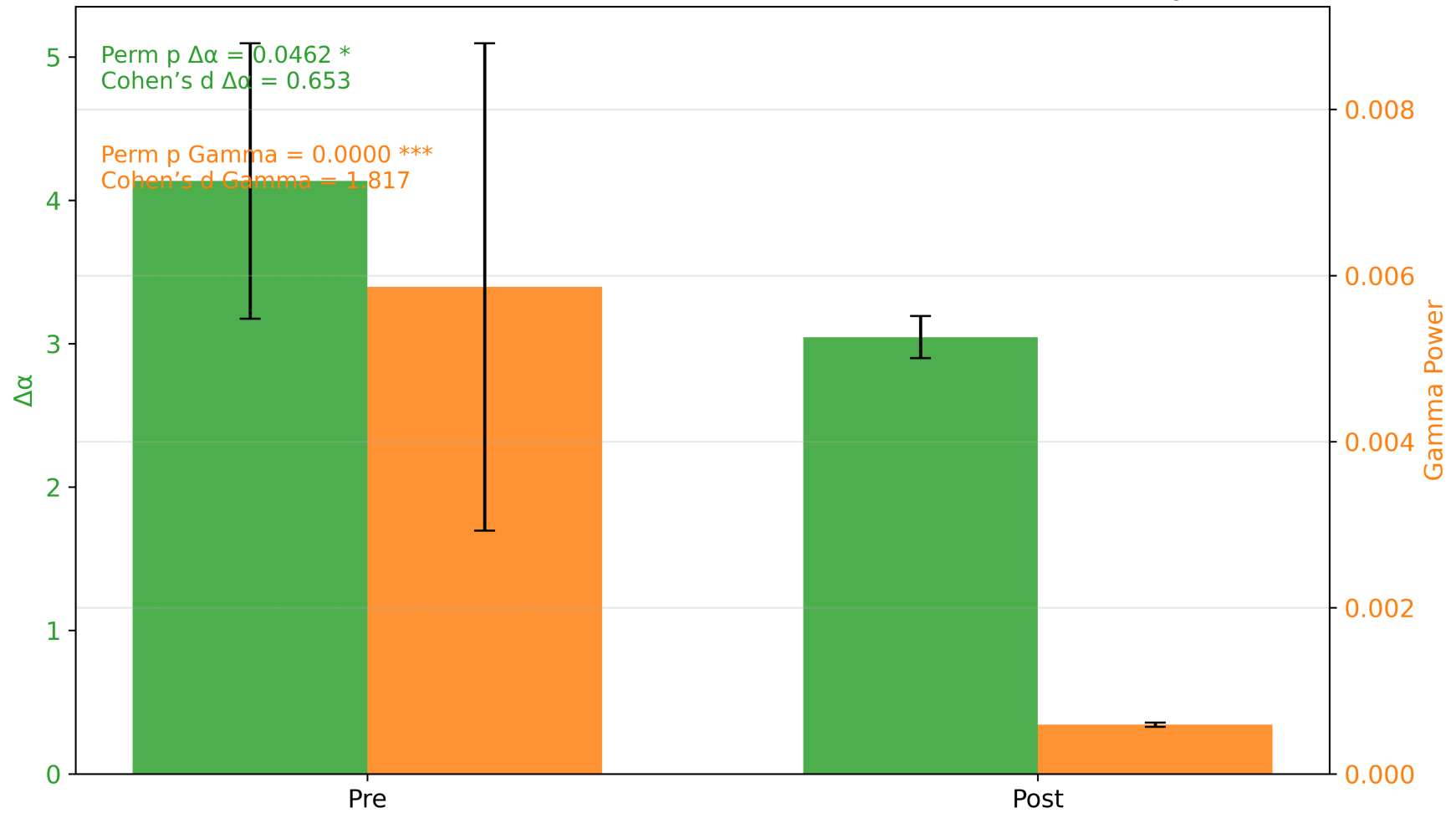


Overlay of $\Delta\alpha$ and multiscale entropy timecourses ($t_{\text{peak}} \approx 22.132812$ s)

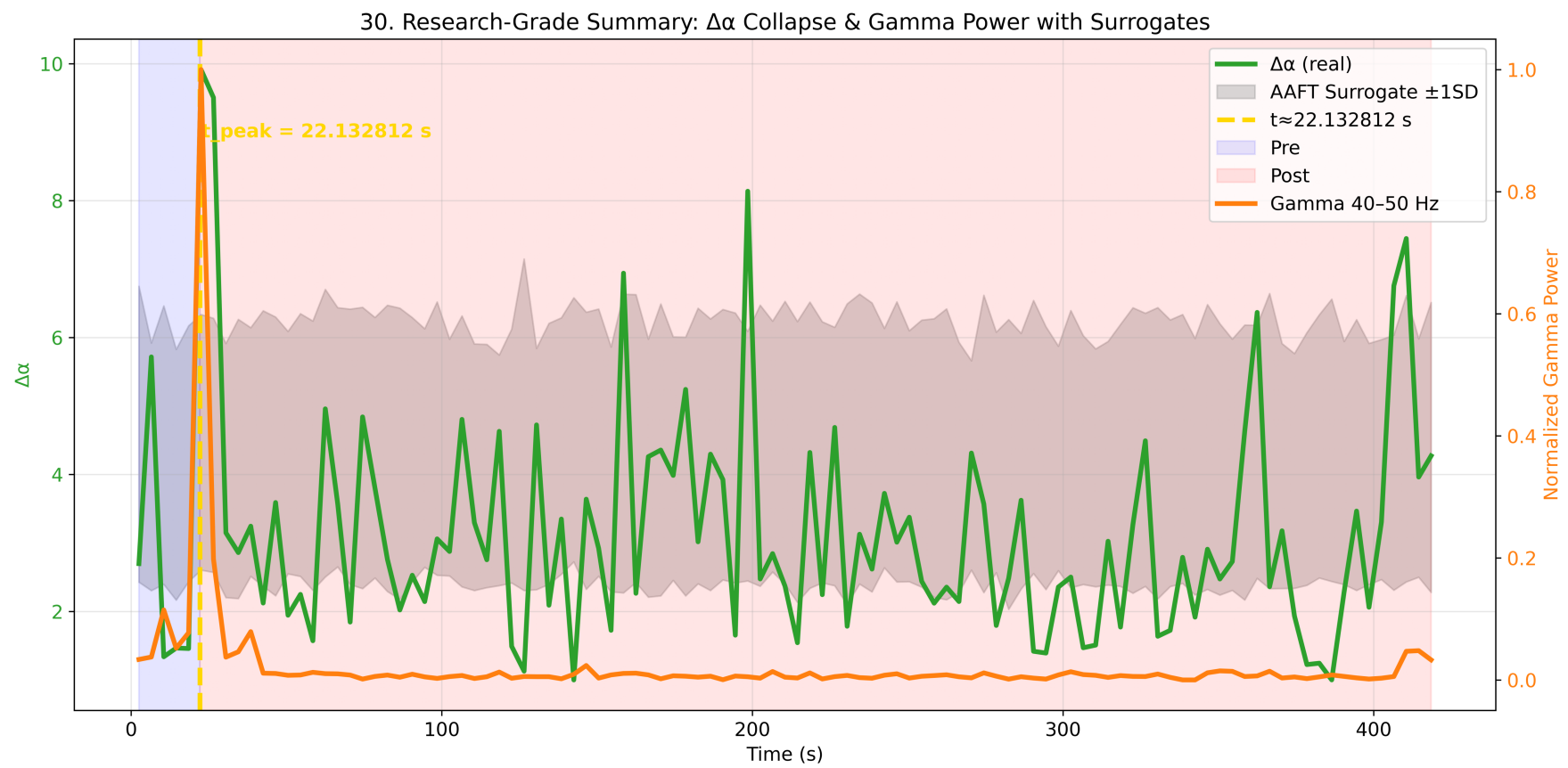


2D density of gamma amplitude on theta phase ($t_{\text{peak}} \approx 22.132812$ s)

29. $\Delta\alpha$ and Gamma Power Pre vs Post $t \approx 41$ s (Main Effect Summary)



$\Delta\alpha$ and gamma power pre vs post $t \approx 22.132812$ s with Cohen's d and p-values



Research-grade summary: $\Delta\alpha$ collapse & gamma power with surrogate band and objective peak at 22.132812 s